

and $m_1 + m_2 + \cdots + m_p = n$. Then

$$f(A) := Zf(J_A)Z^{-1} = Z \operatorname{diag}(f(J_k(\lambda_k)))Z^{-1}, \quad (11.2)$$

where

$$f(J_k(\lambda_k)) := \begin{bmatrix} f(\lambda_k) & f'(\lambda_k) & \cdots & \frac{f^{(m_k-1)}(\lambda_k)}{(m_k-1)!} \\ & f(\lambda_k) & \ddots & \vdots \\ & & \ddots & f'(\lambda_k) \\ & & & f(\lambda_k) \end{bmatrix}. \quad (11.3)$$

- *Polynomial interpolation definition.* Denote by $\lambda_1, \dots, \lambda_s$ the distinct eigenvalues of A and let n_i be the **index** of λ_i , that is, the order of the largest Jordan block in which λ_i appears. Then $f(A) := r(A)$, where r is the unique Hermite interpolating polynomial of degree less than $\sum_{i=1}^s n_i$ that satisfies the interpolation conditions

$$r^{(j)}(\lambda_i) = f^{(j)}(\lambda_i), \quad j = 0: n_i - 1, \quad i = 1:s. \quad (11.4)$$

Note that in both these definitions the derivatives in (11.4) must exist in order for $f(A)$ to be defined. The function f is said to be **defined on the spectrum of A** if all the derivatives in (11.4) exist.

- *Cauchy integral definition.*

$$f(A) := \frac{1}{2\pi i} \int_{\Gamma} f(z)(zI - A)^{-1} dz, \quad (11.5)$$

where f is analytic inside a closed contour Γ that encloses $\sigma(A)$.

When the function f is multivalued and A has a repeated eigenvalue occurring in more than one Jordan block (i.e., A is derogatory), the Jordan canonical form definition has more than one interpretation. Usually, for each occurrence of an eigenvalue in different Jordan blocks the same branch is taken for f and its derivatives. This gives a **primary matrix function**. If different branches are taken for the same eigenvalue in two different Jordan blocks, then a **nonprimary matrix function** is obtained. A nonprimary matrix function is not expressible as a polynomial in the matrix, and if such a function is obtained from the Jordan canonical form definition (11.2) then it depends on the matrix Z . In most applications it is primary matrix functions that are of interest. For the rest of this section $f(A)$ is assumed to be a primary matrix function, unless otherwise stated.

Facts:

Proofs of the facts in this section can be found in one or more of [Hig], [HJ91], or [LT85], unless otherwise stated.

1. The Jordan canonical form and polynomial interpolation definitions are equivalent. Both definitions are equivalent to the Cauchy integral definition when f is analytic.
2. $f(A)$ is a polynomial in A and the coefficients of the polynomial depend on A .
3. $f(A)$ commutes with A .
4. $f(A^T) = f(A)^T$.
5. For any nonsingular X , $f(XAX^{-1}) = Xf(A)X^{-1}$.
6. If A is diagonalizable, with $Z^{-1}AZ = D = \operatorname{diag}(d_1, d_2, \dots, d_n)$, then $f(A) = Zf(D)Z^{-1} = Z \operatorname{diag}(f(d_1), f(d_2), \dots, f(d_n))Z^{-1}$.
7. $f(\operatorname{diag}(A_1, A_2, \dots, A_m)) = \operatorname{diag}(f(A_1), f(A_2), \dots, f(A_m))$.

8. Let f and g be functions defined on the spectrum of A .
- (a) If $h(t) = f(t) + g(t)$, then $h(A) = f(A) + g(A)$.
 - (b) If $h(t) = f(t)g(t)$, then $h(A) = f(A)g(A)$.
9. Let $G(u_1, \dots, u_t)$ be a polynomial in $u_1, \dots, u_t$ and let $f_1, \dots, f_t$ be functions defined on the spectrum of A . If $g(\lambda) = G(f_1(\lambda), \dots, f_t(\lambda))$ takes zero values on the spectrum of A , then $g(A) = G(f_1(A), \dots, f_t(A)) = 0$. For example, $\sin^2(A) + \cos^2(A) = I$, $(A^{1/p})^p = A$, and $e^{iA} = \cos A + i \sin A$.
10. Suppose f has a Taylor series expansion

$$f(z) = \sum_{k=0}^{\infty} a_k (z - \alpha)^k \quad \left(a_k = \frac{f^{(k)}(\alpha)}{k!} \right)$$

with radius of convergence r . If $A \in \mathbb{C}^{n \times n}$, then $f(A)$ is defined and is given by

$$f(A) = \sum_{k=0}^{\infty} a_k (A - \alpha I)^k$$

if and only if each of the distinct eigenvalues $\lambda_1, \dots, \lambda_s$ of A satisfies one of the conditions:

- (a) $|\lambda_i - \alpha| < r$.
 - (b) $|\lambda_i - \alpha| = r$ and the series for $f^{n_i-1}(\lambda)$, where n_i is the index of λ_i , is convergent at the point $\lambda = \lambda_i$, $i = 1:s$.
11. [Dav73], [Des63], [GVL96, Theorem 11.1.3]. Let $T \in \mathbb{C}^{n \times n}$ be upper triangular and suppose that f is defined on the spectrum of T . Then $F = f(T)$ is upper triangular with $f_{ii} = f(t_{ii})$ and

$$f_{ij} = \sum_{(s_0, \dots, s_k) \in S_{ij}} t_{s_0, s_1} t_{s_1, s_2} \dots t_{s_{k-1}, s_k} f[\lambda_{s_0}, \dots, \lambda_{s_k}],$$

where $\lambda_i = t_{ii}$, S_{ij} is the set of all strictly increasing sequences of integers that start at i and end at j , and $f[\lambda_{s_0}, \dots, \lambda_{s_k}]$ is the k th order divided difference of f at $\lambda_{s_0}, \dots, \lambda_{s_k}$.

Examples:

1. For $\lambda_1 \neq \lambda_2$,

$$f\left(\begin{bmatrix} \lambda_1 & \alpha \\ 0 & \lambda_2 \end{bmatrix}\right) = \begin{bmatrix} f(\lambda_1) & \alpha \frac{f(\lambda_2) - f(\lambda_1)}{\lambda_2 - \lambda_1} \\ 0 & f(\lambda_2) \end{bmatrix}.$$

For $\lambda_1 = \lambda_2 = \lambda$,

$$f\left(\begin{bmatrix} \lambda & \alpha \\ 0 & \lambda \end{bmatrix}\right) = \begin{bmatrix} f(\lambda) & \alpha f'(\lambda) \\ 0 & f(\lambda) \end{bmatrix}.$$

2. Compute e^A for the matrix

$$A = \begin{bmatrix} -7 & -4 & -3 \\ 10 & 6 & 4 \\ 6 & 3 & 3 \end{bmatrix}.$$

We have $A = XJ_AX^{-1}$, where $J_A = [0] \oplus \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and

$$X = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix}.$$

Hence, using the Jordan canonical form definition, we have

$$\begin{aligned}
 e^A &= X e_A^J X^{-1} = X \left([1] \oplus \begin{bmatrix} e & e \\ 0 & e \end{bmatrix} \right) X^{-1} \\
 &= \begin{bmatrix} 1 & -1 & -1 \\ -1 & 2 & 0 \\ -1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & e & e \\ 0 & 0 & e \end{bmatrix} \begin{bmatrix} 6 & 3 & 2 \\ 2 & 2 & 1 \\ 2 & 1 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} 6-7e & 3-4e & 2-3e \\ -6+10e & -3+6e & -2+4e \\ -6+6e & -3+3e & -2+3e \end{bmatrix}.
 \end{aligned}$$

3. Compute $\sqrt{A}$ for the matrix in Example 2. To obtain the square root, we use the polynomial interpolation definition. The eigenvalues of A are 0 and 1, with indices 1 and 2, respectively. The unique polynomial r of degree at most 2 satisfying the interpolation conditions $r(0) = f(0)$, $r(1) = f(1)$, $r'(1) = f'(1)$ is

$$r(t) = f(0)(t-1)^2 + t(2-t)f(1) + t(t-1)f'(1).$$

With $f(t) = t^{1/2}$, taking the positive square root, we have $r(t) = t(2-t) + t(t-1)/2$ and, therefore,

$$A^{1/2} = A(2I - A) + A(A - I)/2 = \begin{bmatrix} -6 & -3.5 & -2.5 \\ 8 & 5 & 3 \\ 6 & 3 & 3 \end{bmatrix}.$$

4. Consider the $m_k \times m_k$ Jordan block $J_k(\lambda_k)$ in (11.1). The polynomial satisfying the interpolation conditions (11.4) is

$$r(t) = f(\lambda_k) + (t - \lambda_k)f'(\lambda_k) + \frac{(t - \lambda_k)^2}{2!}f''(\lambda_k) + \cdots + \frac{(t - \lambda_k)^{m_k-1}}{(m_k - 1)!}f^{(m_k-1)}(\lambda_k),$$

which, of course, is the first m_k terms of the Taylor series of f about λ_k . Hence, from the polynomial interpolation definition,

$$\begin{aligned}
 f(J_k(\lambda_k)) &= r(J_k(\lambda_k)) \\
 &= f(\lambda_k)I + (J_k(\lambda_k) - \lambda_k I)f'(\lambda_k) + \frac{(J_k(\lambda_k) - \lambda_k I)^2}{2!}f''(\lambda_k) + \cdots \\
 &\quad + \frac{(J_k(\lambda_k) - \lambda_k I)^{m_k-1}}{(m_k - 1)!}f^{(m_k-1)}(\lambda_k).
 \end{aligned}$$

The matrix $(J_k(\lambda_k) - \lambda_k I)^j$ is zero except for 1s on the j th superdiagonal. This expression for $f(J_k(\lambda_k))$ is, therefore, equal to that in (11.3), confirming the consistency of the first two definitions of $f(A)$.

11.2 Matrix Square Root

Definitions:

Let $A \in \mathbb{C}^{n \times n}$. Any X such that $X^2 = A$ is a **square root** of A .

Facts:

Proofs of the facts in this section can be found in one or more of [Hig], [HJ91], or [LT85], unless otherwise stated.

1. If $A \in \mathbb{C}^{n \times n}$ has no eigenvalues on $\mathbb{R}_0^-$ (the closed negative real axis) then there is a unique square root X of A each of whose eigenvalues is 0 or lies in the open right half-plane, and it is a primary matrix function of A . This is the **principal square root** of A and is written $X = A^{1/2}$. If A is real then $A^{1/2}$ is real. An integral representation is

$$A^{1/2} = \frac{2}{\pi} A \int_0^\infty (t^2 I + A)^{-1} dt.$$

2. A positive (semi)definite matrix $A \in \mathbb{C}^{n \times n}$ has a unique positive (semi)definite square root. (See also Section 8.3.)
3. [CL74] A singular matrix $A \in \mathbb{C}^{n \times n}$ may or may not have a square root. A necessary and sufficient condition for A to have a square root is that in the “ascent sequence” of integers $d_1, d_2, \dots$ defined by

$$d_i = \dim(\ker(A^i)) - \dim(\ker(A^{i-1})),$$

no two terms are the same odd integer.

4. $A \in \mathbb{R}^{n \times n}$ has a real square root if and only if A satisfies the condition in the previous fact and A has an even number of Jordan blocks of each size for every negative eigenvalue.
5. The $n \times n$ identity matrix I_n has 2^n diagonal square roots $\text{diag}(\pm 1)$. Only two of these are primary matrix functions, namely I and $-I$. Nondiagonal but symmetric nonprimary square roots of I_n include any Householder matrix $I - 2\mathbf{v}\mathbf{v}^T/(\mathbf{v}^T\mathbf{v})$ ($\mathbf{v} \neq 0$) and the identity matrix with its columns in reverse order. Nonsymmetric square roots of I_n are easily constructed in the form $XD X^{-1}$, where X is nonsingular but nonorthogonal and $D = \text{diag}(\pm 1) \neq \pm I$.

Examples:

1. The Jordan block $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ has no square root. The matrix

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

has ascent sequence 2, 1, 0, $\dots$ and so does have a square root — for example, the matrix

$$\begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

11.3 Matrix Exponential

Definitions:

The exponential of $A \in \mathbb{C}^{n \times n}$, written e^A or $\exp(A)$, is defined by

$$e^A = I + A + \frac{A^2}{2!} + \cdots + \frac{A^k}{k!} + \cdots.$$

Facts:

Proofs of the facts in this section can be found in one or more of [Hig], [HJ91], or [LT85], unless otherwise stated.

1. $e^{(A+B)t} = e^{At}e^{Bt}$ holds for all t if and only if $AB = BA$.
2. The differential equation in $n \times n$ matrices

$$\frac{dY}{dt} = AY, \quad Y(0) = C, \quad A, Y \in \mathbb{C}^{n \times n},$$

has solution $Y(t) = e^{At}C$.

3. The differential equation in $n \times n$ matrices

$$\frac{dY}{dt} = AY + YB, \quad Y(0) = C, \quad A, B, Y \in \mathbb{C}^{n \times n},$$

has solution $Y(t) = e^{At}Ce^{Bt}$.

4. $A \in \mathbb{C}^{n \times n}$ is unitary if and only if it can be written $A = e^{iH}$, where H is Hermitian. In this representation H can be taken to be Hermitian positive definite.
5. $A \in \mathbb{R}^{n \times n}$ is orthogonal with $\det(A) = 1$ if and only if $A = e^S$ with $S \in \mathbb{R}^{n \times n}$ skew-symmetric.

Examples:

1. Fact 5 is illustrated by the matrix

$$A = \begin{bmatrix} 0 & \alpha \\ -\alpha & 0 \end{bmatrix},$$

for which

$$e^A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}.$$

11.4 Matrix Logarithm

Definitions:

Let $A \in \mathbb{C}^{n \times n}$. Any X such that $e^X = A$ is a **logarithm** of A .

Facts:

Proofs of the facts in this section can be found in one or more of [Hig], [HJ91], or [LT85], unless otherwise stated.

1. If A has no eigenvalues on $\mathbb{R}^-$, then there is a unique logarithm X of A all of whose eigenvalues lie in the strip $\{z : -\pi < \operatorname{Im}(z) < \pi\}$. This is the **principal logarithm** of A and is written $X = \log A$. If A is real, then $\log A$ is real.
2. If $\rho(A) < 1$,

$$\log(I + A) = A - \frac{A^2}{2} + \frac{A^3}{3} - \frac{A^4}{4} + \cdots.$$

3. $A \in \mathbb{R}^{n \times n}$ has a real logarithm if and only if A is nonsingular and A has an even number of Jordan blocks of each size for every negative eigenvalue.
4. $\exp(\log A) = A$ holds when $\log$ is defined on the spectrum of $A \in \mathbb{C}^{n \times n}$. But $\log(\exp(A)) = A$ does not generally hold unless the spectrum of A is restricted.
5. If $A \in \mathbb{C}^{n \times n}$ is nonsingular then $\det(A) = \exp(\operatorname{tr}(\log A))$, where $\log A$ is any logarithm of A .

Examples:

For the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

we have

$$\log(A) = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

11.5 Matrix Sine and Cosine

Definitions:

The sine and cosine of $A \in \mathbb{C}^{n \times n}$ are defined by

$$\begin{aligned} \cos(A) &= I - \frac{A^2}{2!} + \cdots + \frac{(-1)^k}{(2k)!} A^{2k} + \cdots, \\ \sin(A) &= A - \frac{A^3}{3!} + \cdots + \frac{(-1)^k}{(2k+1)!} A^{2k+1} + \cdots. \end{aligned}$$

Facts:

Proofs of the facts in this subsection can be found in one or more of [Hig], [HJ91], or [LT85], unless otherwise stated.

1. $\cos(2A) = 2\cos^2(A) - I$.
2. $\sin(2A) = 2\sin(A)\cos(A)$.
3. $\cos^2(A) + \sin^2(A) = I$.
4. The differential equation

$$\frac{d^2 y}{dt^2} + Ay = 0, \quad y(0) = y_0, \quad y'(0) = y'_0$$

has solution

$$y(t) = \cos(\sqrt{A}t)y_0 + \left(\sqrt{A}\right)^{-1} \sin(\sqrt{A}t)y'_0,$$

where $\sqrt{A}$ denotes any square root of A .

Examples:

1. For

$$A = \begin{bmatrix} 0 & i\alpha \\ i\alpha & 0 \end{bmatrix},$$

we have

$$e^A = \begin{bmatrix} \cos \alpha & i \sin \alpha \\ i \sin \alpha & \cos \alpha \end{bmatrix}.$$

2. For

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -3 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & -1 \end{bmatrix},$$

we have

$$\cos(A) = \cos(1)I, \quad \sin(A) = \begin{bmatrix} \sin(1) & \sin(1) & \sin(1) & \sin(1) \\ 0 & -\sin(1) & -2\sin(1) & -3\sin(1) \\ 0 & 0 & \sin(1) & 3\sin(1) \\ 0 & 0 & 0 & -\sin(1) \end{bmatrix},$$

and $\sin^2(A) = \sin(1)^2 I$, so $\cos(A)^2 + \sin(A)^2 = I$.

11.6 Matrix Sign Function

Definitions:

If $A = ZJ_A Z^{-1} \in \mathbb{C}^{n \times n}$ is a Jordan canonical form arranged so that

$$J_A = \begin{bmatrix} J_A^{(1)} & 0 \\ 0 & J_A^{(2)} \end{bmatrix},$$

where the eigenvalues of $J_A^{(1)} \in \mathbb{C}^{p \times p}$ lie in the open left half-plane and those of $J_A^{(2)} \in \mathbb{C}^{q \times q}$ lie in the open right half-plane, with $p + q = n$, then

$$\text{sign}(A) = Z \begin{bmatrix} -I_p & 0 \\ 0 & I_q \end{bmatrix} Z^{-1}.$$

Alternative formulas are

$$\begin{aligned} \text{sign}(A) &= A(A^2)^{-1/2}, \\ \text{sign}(A) &= \frac{2}{\pi} A \int_0^\infty (t^2 I + A^2)^{-1} dt. \end{aligned} \tag{11.6}$$

If A has any pure imaginary eigenvalues, then $\text{sign}(A)$ is not defined.

Facts:

Proofs of the facts in this section can be found in [Hig].

Let $S = \text{sign}(A)$ be defined. Then

1. $S^2 = I$ (S is involutory).
2. S is diagonalizable with eigenvalues ± 1 .
3. $SA = AS$.
4. If A is real, then S is real.
5. If A is symmetric positive definite, then $\text{sign}(A) = I$.

Examples:

1. For the matrix A in Example 2 of the previous subsection we have $\text{sign}(A) = A$, which follows from (11.6) and the fact that A is involutory.

11.7 Computational Methods for General Functions

Many methods have been proposed for evaluating matrix functions. Three general approaches of wide applicability are outlined here. They have in common that they do not require knowledge of Jordan structure and are suitable for computer implementation. References for this subsection are [GVL96], [Hig].

1. Polynomial and Rational Approximations:

Polynomial approximations

$$p_m(X) = \sum_{k=0}^m b_k X^k, \quad b_k \in \mathbb{C}, \quad X \in \mathbb{C}^{n \times n},$$

to matrix functions can be obtained by truncating or economizing a power series representation, or by constructing a best approximation (in some norm) of a given degree. How to most efficiently evaluate a polynomial at a matrix argument is a nontrivial question. Possibilities include Horner's method, explicit computation of the powers of the matrix, and a method of Paterson and Stockmeyer [GVL96, Sec. 11.2.4], [PS73], which is a combination of these two methods that requires fewer matrix multiplications.

Rational approximations $r_{mk}(X) = p_m(X)q_k(X)^{-1}$ are also widely used, particularly those arising from Padé approximation, which produces rationals matching as many terms of the Taylor series of the function at the origin as possible. The evaluation of rationals at matrix arguments needs careful consideration in order to find the best compromise between speed and accuracy. The main possibilities are

- Evaluating the numerator and denominator polynomials and then solving a multiple right-hand side linear system.
- Evaluating a continued fraction representation (in either top-down or bottom-up order).
- Evaluating a partial fraction representation.

Since polynomials and rationals are typically accurate over a limited range of matrices, practical methods involve a reduction stage prior to evaluating the polynomial or rational.

2. Factorization Methods:

Many methods are based on the property $f(XAX^{-1}) = Xf(A)X^{-1}$. If X can be found such that $B = XAX^{-1}$ has the property that $f(B)$ is easily evaluated, then an obvious method results. When A is diagonalizable, B can be taken to be diagonal, and evaluation of $f(B)$ is trivial. In finite precision arithmetic, though, this approach is reliable only if X is well conditioned, that is, if the condition number $\kappa(X) = \|X\|\|X^{-1}\|$ is not too large. Ideally, X will be unitary, so that in the 2-norm $\kappa_2(X) = 1$. For Hermitian A , or more generally normal A , the spectral decomposition $A = QDQ^*$ with Q unitary and D diagonal is always possible, and if this decomposition can be computed then the formula $f(A) = Qf(D)Q^*$ provides an excellent way of computing $f(A)$.

For general A , if X is restricted to be unitary, then the furthest that A can be reduced is to Schur form: $A = QTQ^*$, where Q is unitary and T upper triangular. This decomposition is computed by the QR algorithm. Computing a function of a triangular matrix is an interesting problem. While Fact 11 of section 11.1 gives an explicit formula for $F = f(T)$, the formula is not practically viable due to its exponential cost in n . Much more efficient is a recurrence of Parlett [Par76]. This is derived by starting with the observation that since F is representable as a polynomial in T , F is upper triangular, with diagonal elements $f(t_{ii})$. The elements in the strict upper triangle are determined by solving the equation $FT = TF$. Parlett's recurrence is:

Algorithm 1. Parlett's recurrence.

```

 $f_{ii} = f(t_{ii}), i = 1:n$ 
for  $j = 2:n$ 
  for  $i = j - 1:-1:1$ 
     $f_{ij} = t_{ij} \frac{f_{ii} - f_{jj}}{t_{ii} - t_{jj}} + \left( \sum_{k=i+1}^{j-1} f_{ik} t_{kj} - t_{ik} f_{kj} \right) / (t_{ii} - t_{jj})$ 
  end
end

```

This recurrence can be evaluated in $2n^3/3$ operations. The recurrence breaks down when $t_{ii} = t_{jj}$ for some $i \neq j$. In this case, T can be regarded as a block matrix $T = (T_{ij})$, with square diagonal blocks, possibly of different sizes. T can be reordered so that no two diagonal blocks have an eigenvalue in common; reordering means applying a unitary similarity transformation to permute the diagonal elements whilst preserving triangularity. Then a block form of the recurrence can be employed. This requires the evaluation of the diagonal blocks $F_{ii} = f(T_{ii})$, where T_{ii} will typically be of small dimension. A general way to obtain F_{ii} is via a Taylor series. The use of the block Parlett recurrence in combination with a Schur decomposition represents the state of the art in evaluation of $f(A)$ for general functions [DH03].

3. Iteration Methods:

Several matrix functions f can be computed by iteration:

$$X_{k+1} = g(X_k), \quad X_0 = A, \quad (11.7)$$

where, for reasons of computational cost, g is usually a polynomial or a rational function. Such an iteration might converge for all A for which f is defined, or just for a subset of such A . A standard means of deriving matrix iterations is to apply Newton's method to an algebraic equation satisfied by $f(A)$. The iterations most used in practice are quadratically convergent, but iterations with higher orders of convergence are known.

4. Contour Integration:

The Cauchy integral definition (11.5) provides a way to compute or approximate $f(A)$ via contour integration. While not suitable as a practical method for all functions or all matrices, this approach can be effective when numerical integration is done over a suitable contour using the repeated trapezium rule, whose high accuracy properties for periodic functions integrated over a whole period are beneficial [DH05], [TW05].

11.8 Computational Methods for Specific Functions

Some methods specialized to particular functions are now outlined. References for this section are [GVL96], [Hig].

1. Matrix Exponential:

A large number of methods have been proposed for the matrix exponential, many of them of pedagogic interest only or of dubious numerical stability. Some of the more computationally useful methods are surveyed in [MVL03]. Probably the best general-purpose method is the scaling and squaring method. In this method an integral power of 2, $\sigma = 2^s$ say, is chosen so that A/σ has norm not too far from 1. The exponential of the scaled matrix is approximated by an $[m/m]$ Padé approximant, $e^{A/2^s} \approx r_{mm}(A/2^s)$, and then s repeated squarings recover an approximation to e^A : $e^A \approx r_{mm}(A/2^s)^{2^s}$. Symmetries in the Padé

approximant permit an efficient evaluation of $r_{mm}(A)$. The scaling and squaring method was originally developed in [MVL78] and [War77], and it is the method employed by MATLAB's `expm` function. How best to choose σ and m is described in [Hig05].

2. Matrix Logarithm:

The (principal) matrix logarithm can be computed using an inverse scaling and squaring method based on the identity $\log A = 2^k \log A^{1/2^k}$, where A is assumed to have no eigenvalues on $\mathbb{R}^-$. Square roots are taken to make $\|A^{1/2^k} - I\|$ small enough that an $[m/m]$ Padé approximant approximates $\log A^{1/2^k}$ sufficiently accurately, for some suitable m . Then $\log A$ is recovered by multiplying by 2^k . To reduce the cost of computing the square roots and evaluating the Padé approximant, a Schur decomposition can be computed initially so that the method works with a triangular matrix. For details, see [CHKL01], [Hig01], or [KL89, App. A].

3. Matrix Cosine and Sine:

A method analogous to the scaling and squaring method for the exponential is the standard method for computing the matrix cosine. The idea is again to scale A to have norm not too far from 1 and then compute a Padé approximant. The difference is that the scaling is undone by repeated use of the double-angle formula $\cos(2A) = 2 \cos^2 A - I$, rather than by repeated squaring. The sine function can be obtained as $\sin(A) = \cos(A - \frac{\pi}{2}I)$. (See [SB80], [HS03], [HH05].)

4. Matrix Square Root:

The most numerically reliable way to compute matrix square roots is via the Schur decomposition, $A = QTQ^*$ [BH83]. Rather than use the Parlett recurrence, a square root U of the upper triangular factor T can be computed by directly solving the equation $U^2 = T$. The choices of sign in the diagonal of U , $u_{ii} = \sqrt{t_{ii}}$, determine which square root is obtained. When A is real, the real Schur decomposition can be used to compute real square roots entirely in real arithmetic [Hig87].

Various iterations exist for computing the principal square root when A has no eigenvalues on $\mathbb{R}^-$. The basic Newton iteration,

$$X_{k+1} = \frac{1}{2}(X_k + X_k^{-1}A), \quad X_0 = A, \quad (11.8)$$

is quadratically convergent, but is numerically unstable unless A is extremely well conditioned and its use is not recommended [Hig86]. Stable alternatives include the Denman–Beavers iteration [DB76]

$$\begin{aligned} X_{k+1} &= \frac{1}{2}(X_k + Y_k^{-1}), & X_0 &= A, \\ Y_{k+1} &= \frac{1}{2}(Y_k + X_k^{-1}), & Y_0 &= I, \end{aligned}$$

for which $\lim_{k \rightarrow \infty} X_k = A^{1/2}$ and $\lim_{k \rightarrow \infty} Y_k = A^{-1/2}$, and the Meini iteration [Mei04]

$$\begin{aligned} Y_{k+1} &= -Y_k Z_k^{-1} Y_k, & Y_0 &= I - A, \\ Z_{k+1} &= Z_k + 2Y_{k+1}, & Z_0 &= 2(I + A), \end{aligned}$$

for which $Y_k \rightarrow 0$ and $Z_k \rightarrow 4A^{1/2}$. Both of these iterations are mathematically equivalent to (11.8) and, hence, are quadratically convergent.

An iteration not involving matrix inverses is the Schulz iteration

$$\begin{aligned} Y_{k+1} &= \frac{1}{2}Y_k(3I - Z_k Y_k), & Y_0 &= A, \\ Z_{k+1} &= \frac{1}{2}(3I - Z_k Y_k)Z_k, & Z_0 &= I, \end{aligned}$$

for which $Y_k \rightarrow A^{1/2}$ and $Z_k \rightarrow A^{-1/2}$ quadratically provided that $\|\text{diag}(A - I, A - I)\| < 1$, where the norm is any consistent matrix norm [Hig97].

5. Matrix Sign Function:

The standard method for computing the matrix sign function is the Newton iteration

$$X_{k+1} = \frac{1}{2}(X_k + X_k^{-1}), \quad X_0 = A,$$

which converges quadratically to $\text{sign}(A)$, provided A has no pure imaginary eigenvalues. In practice, a scaled iteration

$$X_{k+1} = \frac{1}{2}(\mu_k X_k + \mu_k^{-1} X_k^{-1}), \quad X_0 = A$$

is used, where the scale parameters μ_k are chosen to reduce the number of iterations needed to enter the regime where asymptotic quadratic convergence sets in. (See [Bye87], [KL92].)

The Newton–Schulz iteration

$$X_{k+1} = \frac{1}{2}X_k(3I - X_k^2), \quad X_0 = A,$$

involves no matrix inverses, but convergence is guaranteed only for $\|I - A^2\| < 1$.

A Padé family of iterations

$$X_{k+1} = X_k p_{\ell m} (1 - X_k^2) q_{\ell m} (1 - X_k^2)^{-1}, \quad X_0 = A$$

is obtained in [KL91], where $p_{\ell m}(\xi)/q_{\ell m}(\xi)$ is the $[\ell/m]$ Padé approximant to $(1 - \xi)^{-1/2}$. The iteration is globally convergent to $\text{sign}(A)$ for $\ell = m - 1$ and $\ell = m$, and for $\ell \geq m - 1$ is convergent when $\|I - A^2\| < 1$, with order of convergence $\ell + m + 1$ in all cases.

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12

Quadratic, Bilinear, and Sesquilinear Forms

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12.1	Bilinear Forms	12-1
12.2	Symmetric Bilinear Forms	12-3
12.3	Alternating Bilinear Forms	12-5
12.4	φ -Sesquilinear Forms	12-6
12.5	Hermitian Forms	12-7
	References	12-9

Bilinear forms are maps defined on $V \times V$, where V is a vector space, and are linear with respect to each of their variables. There are some similarities between bilinear forms and inner products that are discussed in Chapter 5. Basic properties of bilinear forms, symmetric bilinear forms, and alternating bilinear forms are discussed. The latter two types of forms satisfy additional symmetry conditions.

Quadratic forms are obtained from symmetric bilinear forms by equating the two variables. They are widely used in many areas. A canonical representation of a quadratic form is given when the underlying field is $\mathbb{R}$ or $\mathbb{C}$.

When the field is the complex numbers, it is standard to expect the form to be conjugate linear rather than linear in the second variable; such a form is called sesquilinear. The role of a symmetric bilinear form is played by a Hermitian sesquilinear form. The idea of a sesquilinear form can be generalized to an arbitrary automorphism, encompassing both bilinear and sesquilinear forms as φ -sesquilinear forms, where φ is an automorphism of the field.

Quadratic, bilinear, and φ -sesquilinear forms have applications to classical matrix groups. (See Chapter 67 for more information.)

12.1 Bilinear Forms

It is assumed throughout this section that V is a finite dimensional vector space over a field F .

Definitions:

A **bilinear form** on V is a map f from $V \times V$ into F which satisfies

$$f(a\mathbf{u}_1 + b\mathbf{u}_2, \mathbf{v}) = af(\mathbf{u}_1, \mathbf{v}) + bf(\mathbf{u}_2, \mathbf{v}), \quad \mathbf{u}_1, \mathbf{u}_2, \mathbf{v} \in V, \quad a, b \in F,$$

and

$$f(\mathbf{u}, a\mathbf{v}_1 + b\mathbf{v}_2) = af(\mathbf{u}, \mathbf{v}_1) + bf(\mathbf{u}, \mathbf{v}_2), \quad \mathbf{u}, \mathbf{v}_1, \mathbf{v}_2 \in V, \quad a, b \in F.$$

The space of all bilinear forms on V is denoted $B(V, V, F)$.

Let $\mathcal{B} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ be an ordered basis of V and let $f \in B(V, V, F)$. **The matrix representing f relative to $\mathcal{B}$** is the matrix $A = [a_{ij}] \in F^{n \times n}$ such that $a_{ij} = f(\mathbf{w}_i, \mathbf{w}_j)$.

The **rank** of $f \in B(V, V, F)$, $\text{rank}(f)$, is $\text{rank}(A)$, where A is a matrix representing f relative to an arbitrary ordered basis of V .

$f \in B(V, V, F)$ is **nondegenerate** if its rank is equal to $\dim V$, and **degenerate** if it is not nondegenerate.

Let $A, B \in F^{n \times n}$. B is **congruent** to A if there exists an invertible $P \in F^{n \times n}$ such that $B = P^T A P$.

Let $f, g \in B(V, V, F)$. g is **equivalent** to f if there exists an ordered basis $\mathcal{B}$ of V such that the matrix of g relative to $\mathcal{B}$ is congruent to the matrix of f relative to $\mathcal{B}$.

Let T be a linear operator on V and let $f \in B(V, V, F)$. T **preserves** f if $f(T\mathbf{u}, T\mathbf{v}) = f(\mathbf{u}, \mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$.

Facts:

Let $f \in B(V, V, F)$. The following facts can be found in [HK71, Chap. 10].

1. f is a linear functional in each of its variables when the other variable is held fixed.
2. Let $\mathcal{B} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ be an ordered basis of V and let

$$\mathbf{u} = \sum_{i=1}^n a_i \mathbf{w}_i, \quad \mathbf{v} = \sum_{i=1}^n b_i \mathbf{w}_i.$$

Then,

$$f(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n \sum_{j=1}^n a_i b_j f(\mathbf{w}_i, \mathbf{w}_j).$$

3. Let A denote the matrix representing f relative to $\mathcal{B}$, and let $[\mathbf{u}]_{\mathcal{B}}$ and $[\mathbf{v}]_{\mathcal{B}}$ be the vectors in F^n that are the coordinate vectors of $\mathbf{u}$ and $\mathbf{v}$, respectively, with respect to $\mathcal{B}$. Then $f(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^T A [\mathbf{v}]_{\mathcal{B}}$.
4. Let $\mathcal{B}$ and $\mathcal{B}'$ be ordered bases of V , and P be the matrix whose columns are the $\mathcal{B}$ -coordinates of vectors in $\mathcal{B}'$. Let $f \in B(V, V, F)$. Let A and B denote the matrices representing f relative to $\mathcal{B}$ and $\mathcal{B}'$. Then

$$B = P^T A P.$$

5. The concept of rank of f , as given, is well defined.
6. The set $L = \{\mathbf{v} \in V : f(\mathbf{u}, \mathbf{v}) = 0 \text{ for all } \mathbf{u} \in V\}$ is a subspace of V and $\text{rank}(f) = \dim V - \dim L$. In particular, f is nondegenerate if and only if $L = \{0\}$.
7. Suppose that $\dim V = n$. The space $B(V, V, F)$ is a vector space over F under the obvious addition of two bilinear forms and multiplication of a bilinear form by a scalar. Moreover, $B(V, V, F)$ is isomorphic to $F^{n \times n}$.
8. Congruence is an equivalence relation on $F^{n \times n}$.
9. Let $f \in B(V, V, F)$ be nondegenerate. Then the set of all linear operators on V , which preserve f , is a group under the operation of composition.

Examples:

1. Let $A \in F^{n \times n}$. The map $f : F^n \times F^n \rightarrow F$ defined by

$$f(\mathbf{u}, \mathbf{v}) = \mathbf{u}^T A \mathbf{v} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} u_i v_j, \quad \mathbf{u}, \mathbf{v} \in F^n,$$

is a bilinear form. Since $f(\mathbf{e}_i, \mathbf{e}_j) = a_{ij}$, $i, j = 1, 2, \dots, n$, f is represented in the standard basis of F^n by A . It follows that $\text{rank}(f) = \text{rank}(A)$, and f is nondegenerate if and only if A is invertible.

2. Let $C \in F^{m \times m}$ and $\text{rank}(C) = k$. The map $f : F^{m \times n} \times F^{m \times n} \rightarrow F$ defined by $f(A, B) = \text{tr}(A^T C B)$ is a bilinear form. This follows immediately from the basic properties of the trace function. To compute $\text{rank}(f)$, let L be defined as in Fact 6, that is, $L = \{B \in F^{m \times n} : \text{tr}(A^T C B) = 0 \text{ for all } A \in F^{m \times n}\}$. It follows that $L = \{B \in F^{m \times n} : C B = 0\}$, which implies that $\dim L = n(m - k)$. Hence, $\text{rank}(f) = mn - n(m - k) = kn$. In particular, f is nondegenerate if and only if C is invertible.
3. Let $\mathbb{R}[x; n]$ denote the space of all real polynomials of the form $\sum_{i=0}^n a_i x^i$. Then $f(p(x), q(x)) = p(0)q(0) + p(1)q(1) + p(2)q(2)$ is a bilinear form on $\mathbb{R}[x; n]$. It is nondegenerate if $n = 2$ and degenerate if $n \geq 3$.

12.2 Symmetric Bilinear Forms

It is assumed throughout this section that V is a finite dimensional vector space over a field F .

Definitions:

Let $f \in B(V, V, F)$. Then f is **symmetric** if $f(\mathbf{u}, \mathbf{v}) = f(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in V$.

Let f be a symmetric bilinear form on V , and let $\mathbf{u}, \mathbf{v} \in V$; $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal with respect to f** if $f(\mathbf{u}, \mathbf{v}) = 0$.

Let f be a symmetric bilinear form on V . The **quadratic form corresponding to f** is the map $g : V \rightarrow F$ defined by $g(\mathbf{v}) = f(\mathbf{v}, \mathbf{v})$, $\mathbf{v} \in V$.

A symmetric bilinear form f on a real vector space V is **positive semidefinite (positive definite)** if $f(\mathbf{v}, \mathbf{v}) \geq 0$ for all $\mathbf{v} \in V$ ($f(\mathbf{v}, \mathbf{v}) > 0$ for all $0 \neq \mathbf{v} \in V$).

f is **negative semidefinite (negative definite)** if $-f$ is positive semidefinite (positive definite).

The **signature** of a real symmetric matrix A is the integer $\pi - \nu$, where (π, ν, δ) is the inertia of A . (See Section 8.3.)

The **signature** of a real symmetric bilinear form is the signature of a matrix representing the form relative to some basis.

Facts:

Additional facts about real symmetric matrices can be found in Chapter 8. Except where another reference is provided, the following facts can be found in [Coh89, Chap. 8], [HJ85, Chap. 4], or [HK71, Chap. 10].

1. A positive definite bilinear form is nondegenerate.
2. An inner product on a real vector space is a positive definite symmetric bilinear form. Conversely, a positive definite symmetric bilinear form on a real vector space is an inner product.
3. Let $\mathcal{B}$ be an ordered basis of V and let $f \in B(V, V, F)$. Let A be the matrix representing f relative to $\mathcal{B}$. Then f is symmetric if and only if A is a symmetric matrix, that is, $A = A^T$.
4. Let f be a symmetric bilinear form on V and let g be the quadratic form corresponding to f . Suppose that the characteristic of F is not 2. Then f can be recovered from g :

$$f(\mathbf{u}, \mathbf{v}) = \frac{1}{2}[g(\mathbf{u} + \mathbf{v}) - g(\mathbf{u}) - g(\mathbf{v})] \quad \text{for all } \mathbf{u}, \mathbf{v} \in V.$$

5. Let f be a symmetric bilinear form on V and suppose that the characteristic of F is not 2. Then there exists an ordered basis $\mathcal{B}$ of V such that the matrix representing f relative to it is diagonal; i.e., if $A \in F^{n \times n}$ is a symmetric matrix, then A is congruent to a diagonal matrix.
6. Suppose that V is a complex vector space and f is a symmetric bilinear form on V . Let $r = \text{rank}(f)$. Then there is an ordered basis $\mathcal{B}$ of V such that the matrix representing f relative to $\mathcal{B}$ is $I_r \oplus \mathbf{0}$. In matrix language, this fact states that if $A \in \mathbb{C}^{n \times n}$ is symmetric with $\text{rank}(A) = r$, then it is congruent to $I_r \oplus \mathbf{0}$.
7. The only invariant of $n \times n$ complex symmetric matrices under congruence is the rank.
8. Two complex $n \times n$ symmetric matrices are congruent if and only if they have the same rank.

9. (*Sylvester's law of inertia for symmetric bilinear forms*) Suppose that V is a real vector space and f is a symmetric bilinear form on V . Then there is an ordered basis $\mathcal{B}$ of V such that the matrix representing f relative to it has the form $I_\pi \oplus -I_\nu \oplus \mathbf{0}_\delta$. Moreover, π , ν , and δ do not depend on the choice of $\mathcal{B}$, but only on f .
10. (*Sylvester's law of inertia for matrices*) If $A \in \mathbb{R}^{n \times n}$ is symmetric, then A is congruent to the diagonal matrix $D = I_\pi \oplus -I_\nu \oplus \mathbf{0}_\delta$, where $(\pi, \nu, \delta) = \text{in}(A)$.
11. There are exactly two invariants of $n \times n$ real symmetric matrices under congruence, namely the rank and the signature.
12. Two real $n \times n$ symmetric matrices are congruent if and only if they have the same rank and the same signature.
13. The signature of a real symmetric bilinear form is well defined.
14. Two real symmetric bilinear forms are equivalent if and only if they have the same rank and the same signature.
15. [Hes68] Let $n \geq 3$ and let $A, B \in \mathbb{R}^{n \times n}$ be symmetric. Suppose that $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}^T A \mathbf{x} = \mathbf{x}^T B \mathbf{x} = 0 \Rightarrow \mathbf{x} = \mathbf{0}$. Then $\exists \mathbf{a}, \mathbf{b} \in \mathbb{R}$ such that $\mathbf{a}A + \mathbf{b}B$ is positive definite.
16. The group of linear operators preserving the form $f(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n u_i v_i$ on $\mathbb{R}^n$ is the real n -dimensional orthogonal group, while the group preserving the same form on $\mathbb{C}^n$ is the complex n -dimensional orthogonal group.

Examples:

1. Consider Example 1 in section 12.1. The map f is a symmetric bilinear form if and only if $A = A^T$. The quadratic form g corresponding to f is given by

$$g(\mathbf{u}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} u_i u_j, \quad \mathbf{u} \in F^n.$$

2. Consider Example 2 in section 12.1. The map f is a symmetric bilinear form if and only if $C = C^T$.
3. The symmetric bilinear form $f_{\mathbf{a}}$ on $\mathbb{R}^2$ given by

$$f_{\mathbf{a}}(\mathbf{u}, \mathbf{v}) = u_1 v_1 - 2u_1 v_2 - 2u_2 v_1 + a u_2 v_2, \quad \mathbf{u}, \mathbf{v} \in \mathbb{R}^2, \quad a \in \mathbb{R} \text{ is a parameter,}$$

is an inner product on $\mathbb{R}^2$ if and only if $a > 4$.

4. Since we consider in this article only finite dimensional vector spaces, let V be any finite dimensional subspace of $C[0, 1]$, the space of all real valued, continuous functions on $[0, 1]$. Then the map $f : V \times V \rightarrow \mathbb{R}$ defined by

$$f(\mathbf{u}, \mathbf{v}) = \int_0^1 t^3 u(t) v(t) dt, \quad \mathbf{u}, \mathbf{v} \in V,$$

is a symmetric bilinear form on V .

Applications:

1. Conic sections: Consider the set of points (x_1, x_2) in $\mathbb{R}^2$, which satisfy the equation

$$a x_1^2 + b x_1 x_2 + c x_2^2 + d x_1 + e x_2 + f = 0,$$

where $a, b, c, d, e, f \in \mathbb{R}$. The solution set is a conic section, namely an ellipse, hyperbola, parabola, or a degenerate form of those. The analysis of this equation depends heavily on the quadratic form

$a x_1^2 + b x_1 x_2 + c x_2^2$, which is represented in the standard basis of $\mathbb{R}^2$ by $A = \begin{bmatrix} a & b/2 \\ b/2 & c \end{bmatrix}$. If the

solution of the quadratic equation above represents a nondegenerate conic section, then its type is determined by the sign of $4ac - b^2$. More precisely, the conic is an ellipse, hyperbola, or parabola if $4ac - b^2$ is positive, negative, or zero, respectively.

2. Theory of small oscillations: Suppose a mechanical system undergoes small oscillations about an equilibrium position. Let $x_1, x_2, \dots, x_n$ denote the coordinates of the system, and let $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$. Then the kinetic energy of the system is given by a quadratic form (in the velocities $\dot{x}_1, \dot{x}_2, \dots, \dot{x}_n$) $\frac{1}{2} \dot{\mathbf{x}}^T A \dot{\mathbf{x}}$, where A is a positive definite matrix. If $\mathbf{x} = 0$ is the equilibrium position, then the potential energy of the system is given by another quadratic form $\frac{1}{2} \mathbf{x}^T B \mathbf{x}$, where $B = B^T$. The equations of motion are $A \ddot{\mathbf{x}} + B \mathbf{x} = 0$. It is known that A and B can be simultaneously diagonalized, that is, there exists an invertible $P \in \mathbb{R}^{n \times n}$ such that $P^T A P$ and $P^T B P$ are diagonal matrices. This can be used to obtain the solution of the system.

12.3 Alternating Bilinear Forms

It is assumed throughout this section that V is a finite dimensional vector space over a field F .

Definitions:

Let $f \in B(V, V, F)$. Then f is **alternating** if $f(\mathbf{v}, \mathbf{v}) = 0$ for all $\mathbf{v} \in V$. f is **antisymmetric** if $f(\mathbf{u}, \mathbf{v}) = -f(\mathbf{v}, \mathbf{u})$ for all $\mathbf{u}, \mathbf{v} \in V$.

Let $A \in F^{n \times n}$. Then A is **alternating** if $a_{ii} = 0$, $i = 1, 2, \dots, n$ and $a_{ji} = -a_{ij}$, $1 \leq i < j \leq n$.

Facts:

The following facts can be found in [Coh89, Chap. 8], [HK71, Chap. 10], or [Lan99, Chap. 15].

1. Let $f \in B(V, V, F)$ be alternating. Then f is antisymmetric because for all $\mathbf{u}, \mathbf{v} \in V$,

$$f(\mathbf{u}, \mathbf{v}) + f(\mathbf{v}, \mathbf{u}) = f(\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v}) - f(\mathbf{u}, \mathbf{u}) - f(\mathbf{v}, \mathbf{v}) = 0.$$

The converse is true if the characteristic of F is not 2.

2. Let $A \in F^{n \times n}$ be an alternating matrix. Then $A^T = -A$. The converse is true if the characteristic of F is not 2.
3. Let $\mathcal{B}$ be an ordered basis of V and let $f \in B(V, V, F)$. Let A be the matrix representing f relative to $\mathcal{B}$. Then f is alternating if and only if A is an alternating matrix.
4. Let f be an alternating bilinear form on V and let $r = \text{rank}(f)$. Then r is even and there exists an ordered basis $\mathcal{B}$ of V such that the matrix representing f relative to it has the form

$$\underbrace{\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \oplus \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \oplus \cdots \oplus \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}}_{r/2 - \text{times}} \oplus \mathbf{0}.$$

There is an ordered basis $\mathcal{B}_1$ where f is represented by the matrix $\begin{bmatrix} \mathbf{0} & I_{r/2} \\ -I_{r/2} & \mathbf{0} \end{bmatrix} \oplus \mathbf{0}$.

5. Let $f \in B(V, V, F)$ and suppose that the characteristic of F is not 2. Define:

$$f_1 : V \times V \rightarrow F \quad \text{by} \quad f_1(\mathbf{u}, \mathbf{v}) = \frac{1}{2} [f(\mathbf{u}, \mathbf{v}) + f(\mathbf{v}, \mathbf{u})], \quad \mathbf{u}, \mathbf{v} \in V,$$

$$f_2 : V \times V \rightarrow F \quad \text{by} \quad f_2(\mathbf{u}, \mathbf{v}) = \frac{1}{2} [f(\mathbf{u}, \mathbf{v}) - f(\mathbf{v}, \mathbf{u})], \quad \mathbf{u}, \mathbf{v} \in V.$$

Then f_1 (f_2) is a symmetric (alternating) bilinear form on V , and $f = f_1 + f_2$. Moreover, this representation of f as a sum of a symmetric and an alternating bilinear form is unique.

6. Let $A \in F^{n \times n}$ be an alternating matrix and suppose that A is invertible. Then n is even and A is congruent to the matrix $\begin{bmatrix} \mathbf{0} & I_{n/2} \\ -I_{n/2} & \mathbf{0} \end{bmatrix}$, so $\det(A)$ is a square in F . There exists a polynomial in $n(n-1)/2$ variables, called the **Pfaffian**, such that $\det(A) = a^2$, where $a \in F$ is obtained by substituting into the Pfaffian the entries of A above the main diagonal for the indeterminates.

7. Let f be an alternating nondegenerate bilinear form on V . Then $\dim V = 2m$ for some positive integer m . The group of all linear operators on V that preserve f is the symplectic group.

Examples:

1. Consider Example 1 in section 12.1. The map f is alternating if and only if the matrix A is an alternating matrix.
2. Consider Example 2 in section 12.1. The map f is alternating if and only if C is an alternating matrix.
3. Let $C \in F^{n \times n}$. Define $f : F^{n \times n} \rightarrow F^{n \times n}$ by $f(A, B) = \text{tr}(ACB - BCA)$. Then f is alternating.

12.4 φ -Sesquilinear Forms

This section generalizes Section 12.1, and is consequently very similar. This generalization is required by applications to matrix groups (see Chapter 67), but for most purposes such generality is not required, and the simpler discussion of bilinear forms in Section 12.1 is preferred. It is assumed throughout this section that V is a finite dimensional vector space over a field F and φ is an automorphism of F .

Definitions:

A φ -**sesquilinear form** on V is a map $f : V \times V \rightarrow F$, which is linear as a function in the first variable and φ -semilinear in the second, i.e.,

$$f(a\mathbf{u}_1 + b\mathbf{u}_2, \mathbf{v}) = af(\mathbf{u}_1, \mathbf{v}) + bf(\mathbf{u}_2, \mathbf{v}), \quad \mathbf{u}_1, \mathbf{u}_2, \mathbf{v} \in V, \quad a, b \in F,$$

and

$$f(\mathbf{u}, a\mathbf{v}_1 + b\mathbf{v}_2) = \varphi(a)f(\mathbf{u}, \mathbf{v}_1) + \varphi(b)f(\mathbf{u}, \mathbf{v}_2), \quad \mathbf{u}, \mathbf{v}_1, \mathbf{v}_2 \in V, \quad a, b \in F.$$

In the case $F = \mathbb{C}$ and φ is complex conjugation, a φ -sesquilinear form is called a **sesquilinear form**. The space of all φ -sesquilinear forms on V is denoted $B(V, V, F, \varphi)$.

Let $\mathcal{B} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ be an ordered basis of V and let $f \in B(V, V, F, \varphi)$. The **matrix representing f relative to $\mathcal{B}$** is the matrix $A = [a_{ij}] \in F^{n \times n}$ such that $a_{ij} = f(\mathbf{w}_i, \mathbf{w}_j)$.

The **rank** of $f \in B(V, V, F, \varphi)$, $\text{rank}(f)$, is $\text{rank}(A)$, where A is a matrix representing f relative to an arbitrary ordered basis of V .

$f \in B(V, V, F, \varphi)$ is **nondegenerate** if its rank is equal to $\dim V$, and **degenerate** if it is not nondegenerate.

Let $A = [a_{ij}] \in F^{n \times n}$. $\varphi(A)$ is the $n \times n$ matrix whose i, j -entry is $\varphi(a_{ij})$.

Let $A, B \in F^{n \times n}$. B is **φ -congruent** to A if there exists an invertible $P \in F^{n \times n}$ such that $B = P^T A \varphi(P)$.

Let $f, g \in B(V, V, F, \varphi)$. g is **φ -equivalent** to f if there exists an ordered basis $\mathcal{B}$ of V such that the matrix of g relative to $\mathcal{B}$ is φ -congruent to the matrix of f relative to $\mathcal{B}$.

Let T be a linear operator on V and let $f \in B(V, V, F, \varphi)$. T **preserves f** if $f(T\mathbf{u}, T\mathbf{v}) = f(\mathbf{u}, \mathbf{v})$ for all $\mathbf{u}, \mathbf{v} \in V$.

Facts:

Let $f \in B(V, V, F, \varphi)$. The following facts can be obtained by obvious generalizations of the proofs of the corresponding facts in section 12.1; see that section for references.

1. A bilinear form is a φ -sesquilinear form with the automorphism being the identity map.
2. Let $\mathcal{B} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ be an ordered basis of V and let

$$\mathbf{u} = \sum_{i=1}^n a_i \mathbf{w}_i, \quad \mathbf{v} = \sum_{i=1}^n b_i \mathbf{w}_i.$$

Then,

$$f(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n \sum_{j=1}^n a_i \varphi(b_j) f(\mathbf{w}_i, \mathbf{w}_j).$$

3. Let A denote the matrix representing the φ -sesquilinear f relative to $\mathcal{B}$, and let $[\mathbf{u}]_{\mathcal{B}}$ and $[\mathbf{v}]_{\mathcal{B}}$ be the vectors in F^n , which are the coordinate vectors of $\mathbf{u}$ and $\mathbf{v}$, respectively, with respect to $\mathcal{B}$. Then $f(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^T A \varphi([\mathbf{v}]_{\mathcal{B}})$.
4. Let $\mathcal{B}$ and $\mathcal{B}'$ be ordered bases of V , and P be the matrix whose columns are the $\mathcal{B}$ -coordinates of vectors in $\mathcal{B}'$. Let $f \in B(V, V, F, \varphi)$. Let A and B denote the matrices representing f relative to $\mathcal{B}$ and $\mathcal{B}'$. Then

$$B = P^T A \varphi(P).$$

5. The concept of rank of f , as given, is well defined.
6. The set $L = \{\mathbf{v} \in V : f(\mathbf{u}, \mathbf{v}) = 0 \text{ for all } \mathbf{u} \in V\}$ is a subspace of V and $\text{rank}(f) = \dim V - \dim L$. In particular, f is nondegenerate if and only if $L = \{0\}$.
7. Suppose that $\dim V = n$. The space $B(V, V, F, \varphi)$ is a vector space over F under the obvious addition of two φ -sesquilinear forms and multiplication of a φ -sesquilinear form by a scalar. Moreover, $B(V, V, F, \varphi)$ is isomorphic to $F^{n \times n}$.
8. φ -Congruence is an equivalence relation on $F^{n \times n}$.
9. Let $f \in B(V, V, F, \varphi)$ be nondegenerate. Then the set of all linear operators on V which preserve f is a group under the operation of composition.

Examples:

1. Let $F = \mathbb{Q}(\sqrt{5}) = \{a + b\sqrt{5} : a, b \in \mathbb{Q}\}$ and $\varphi(a + b\sqrt{5}) = a - b\sqrt{5}$. Define the φ -sesquilinear form f on F^2 by $f(\mathbf{u}, \mathbf{v}) = \mathbf{u}^T \varphi(\mathbf{v})$. $f([1 + \sqrt{5}, 3]^T, [-2\sqrt{5}, -1 + \sqrt{5}]^T) = (1 + \sqrt{5})(2\sqrt{5}) + 3(-1 - \sqrt{5}) = 7 - \sqrt{5}$.

The matrix of f with respect to the standard basis is the identity matrix, $\text{rank } f = 2$, and f is nondegenerate.

2. Let $A \in F^{n \times n}$. The map $f : F^n \times F^n \rightarrow F$ defined by

$$f(\mathbf{u}, \mathbf{v}) = \mathbf{u}^T A \varphi(\mathbf{v}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} u_i \varphi(v_j), \quad \mathbf{u}, \mathbf{v} \in F^n,$$

is a φ -sesquilinear form. Since $f(\mathbf{e}_i, \mathbf{e}_j) = a_{ij}$, $i, j = 1, 2, \dots, n$, f is represented in the standard basis of F^n by A . It follows that $\text{rank}(f) = \text{rank}(A)$, and f is nondegenerate if and only if A is invertible.

12.5 Hermitian Forms

This section closely resembles the results related to symmetric bilinear forms on real vector spaces. We assume here that V is a finite dimensional complex vector space.

Definitions:

A **Hermitian form** on V is a map $f : V \times V \rightarrow \mathbb{C}$, which satisfies

$$f(a\mathbf{u}_1 + b\mathbf{u}_2, \mathbf{v}) = af(\mathbf{u}_1, \mathbf{v}) + bf(\mathbf{u}_2, \mathbf{v}), \quad \mathbf{u}, \mathbf{v} \in V, \quad a, b \in \mathbb{C},$$

and

$$f(\mathbf{v}, \mathbf{u}) = \overline{f(\mathbf{u}, \mathbf{v})}, \quad \mathbf{u}, \mathbf{v} \in V.$$

A Hermitian form f on V is **positive semidefinite** (**positive definite**) if $f(\mathbf{v}, \mathbf{v}) \geq 0$ for all $\mathbf{v} \in V$ ($f(\mathbf{v}, \mathbf{v}) > 0$ for all $0 \neq \mathbf{v} \in V$).

f is **negative semidefinite** (**negative definite**) if $-f$ is positive semidefinite (positive definite).

The **signature** of a Hermitian matrix A is the integer $\pi - \nu$, where (π, ν, δ) is the inertia of A . (See Section 8.3.)

The **signature** of a Hermitian form is the signature of a matrix representing the form.

Let $A, B \in \mathbb{C}^{n \times n}$. B is ***congruent** to A if there exists an invertible $S \in \mathbb{C}^{n \times n}$ such that $B = S^*AS$ (where S^* denotes the Hermitian adjoint of S).

Let f, g be Hermitian forms on a finite dimensional complex vector space V . g is ***equivalent** to f if there exists an ordered basis $\mathcal{B}$ of V such that the matrix of g relative to $\mathcal{B}$ is *congruent to the matrix of f relative to $\mathcal{B}$.

Facts:

Except where another reference is provided, the following facts can be found in [Coh89, Chap. 8], [HJ85, Chap. 4], or [Lan99, Chap. 15]. Let f be a Hermitian form on V .

1. A Hermitian form is sesquilinear.
2. A positive definite Hermitian form is nondegenerate.
3. f is a linear functional in the first variable and conjugate linear in the second variable, that is,

$$f(\mathbf{u}, a\mathbf{v}_1 + b\mathbf{v}_2) = \bar{a}f(\mathbf{u}, \mathbf{v}_1) + \bar{b}f(\mathbf{u}, \mathbf{v}_2).$$

4. $f(\mathbf{v}, \mathbf{v}) \in \mathbb{R}$ for all $\mathbf{v} \in V$.
5. An inner product on a complex vector space is a positive definite Hermitian form. Conversely, a positive definite Hermitian form on a complex vector space is an inner product.
6. (*Polarization formula*)

$$4f(\mathbf{u}, \mathbf{v}) = f(\mathbf{u} + \mathbf{v}, \mathbf{u} + \mathbf{v}) - f(\mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v}) + if(\mathbf{u} + i\mathbf{v}, \mathbf{u} + i\mathbf{v}) - if(\mathbf{u} - i\mathbf{v}, \mathbf{u} - i\mathbf{v}).$$

7. Let $\mathcal{B} = (\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n)$ be an ordered basis of V and let

$$\mathbf{u} = \sum_{i=1}^n a_i \mathbf{w}_i, \quad \mathbf{v} = \sum_{i=1}^n b_i \mathbf{w}_i.$$

Then

$$f(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n \sum_{j=1}^n a_i \bar{b}_j f(\mathbf{w}_i, \mathbf{w}_j).$$

8. Let A denote the matrix representing f relative to the basis $\mathcal{B}$. Then

$$f(\mathbf{u}, \mathbf{v}) = [\mathbf{u}]_{\mathcal{B}}^T A [\bar{\mathbf{v}}]_{\mathcal{B}}.$$

9. The matrix representing a Hermitian form f relative to any basis of V is a Hermitian matrix.
10. Let A, B be matrices that represent f relative to bases $\mathcal{B}$ and $\mathcal{B}'$ of V , respectively. Then B is *congruent to A .
11. (*Sylvester's law of inertia for Hermitian forms*, cf. 12.2) There exists an ordered basis $\mathcal{B}$ of V such that the matrix representing f relative to it has the form

$$I_{\pi} \oplus -I_{\nu} \oplus \mathbf{0}_{\delta}.$$

Moreover, π , ν , and δ depend only on f and not on the choice of $\mathcal{B}$.

12. (*Sylvester's law of inertia for Hermitian matrices*, cf. 12.2) If $A \in \mathbb{C}^{n \times n}$ is a Hermitian matrix, then A is *congruent to the diagonal matrix $D = I_{\pi} \oplus -I_{\nu} \oplus \mathbf{0}_{\delta}$, where $(\pi, \nu, \delta) = \text{in}(A)$.

13. There are exactly two invariants of $n \times n$ Hermitian matrices under $*$ -congruence, namely the rank and the signature.
14. Two Hermitian $n \times n$ matrices are $*$ -congruent if and only if they have the same rank and the same signature.
15. The signature of a Hermitian form is well-defined.
16. Two Hermitian forms are $*$ -equivalent if and only if they have the same rank and the same signature.
17. [HJ91, Theorem 1.3.5] Let $A, B \in \mathbb{C}^{n \times n}$ be Hermitian matrices. Suppose that $\mathbf{x} \in \mathbb{C}^n$, $\mathbf{x}^* A \mathbf{x} = \mathbf{x}^* B \mathbf{x} = 0 \Rightarrow \mathbf{x} = 0$. Then $\exists \mathbf{a}, \mathbf{b} \in \mathbb{R}$ such that $\mathbf{a}A + \mathbf{b}B$ is positive definite. This fact can be obtained from [HJ91], where it is stated in a slightly different form, using the decomposition of every square, complex matrix as a sum of a Hermitian matrix and a skew-Hermitian matrix.
18. The group of linear operators preserving the Hermitian form $f(u, v) = \sum_{i=1}^n u_i \bar{v}_i$ on $\mathbb{C}^n$ is the n -dimensional unitary group.

Examples:

1. Let $A \in \mathbb{C}^{n \times n}$ be a Hermitian matrix. The map $f: \mathbb{C}^n \times \mathbb{C}^n \rightarrow \mathbb{C}$ defined by $f(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n \sum_{j=1}^n a_{ij} u_i \bar{v}_j$ is a Hermitian form on $\mathbb{C}^n$.
2. Let $\psi_1, \psi_2, \dots, \psi_k$ be linear functionals on V , and let $a_1, a_2, \dots, a_k \in \mathbb{R}$. Then the map $f: V \times V \rightarrow \mathbb{C}$ defined by $f(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^k a_i \psi_i(\mathbf{u}) \overline{\psi_i(\mathbf{v})}$ is a Hermitian form on V .
3. Let $H \in \mathbb{C}^{n \times n}$ be a Hermitian matrix.
The map $f: \mathbb{C}^{n \times n} \times \mathbb{C}^{n \times n} \rightarrow \mathbb{C}$ defined by $f(A, B) = \text{tr}(AHB^*)$ is a Hermitian form.

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13

Multilinear Algebra

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13.1	Multilinear Maps.....	13-1
13.2	Tensor Products.....	13-3
13.3	Rank of a Tensor: Decomposable Tensors	13-7
13.4	Tensor Product of Linear Maps	13-8
13.5	Symmetric and Antisymmetric Maps	13-10
13.6	Symmetric and Grassmann Tensors	13-12
13.7	The Tensor Multiplication, the Alt Multiplication, and the Sym Multiplication.....	13-17
13.8	Associated Maps	13-19
13.9	Tensor Algebras	13-20
13.10	Tensor Product of Inner Product Spaces.....	13-22
13.11	Orientation and Hodge Star Operator	13-24
	References	13-26

13.1 Multilinear Maps

Unless otherwise stated, within this section V , U , and W as well as these letters with subscripts, superscripts, or accents, are finite dimensional vector spaces over a field F of characteristic zero.

Definitions:

A map φ from $V_1 \times \cdots \times V_m$ into U is a **multilinear map** (**m -linear map**) if it is linear on each coordinate, i.e., for every $\mathbf{v}_i, \mathbf{v}'_i \in V_i$, $i = 1, \dots, m$ and for every $a \in F$ the following conditions hold:

- (a) $\varphi(\mathbf{v}_1, \dots, \mathbf{v}_i + \mathbf{v}'_i, \dots, \mathbf{v}_m) = \varphi(\mathbf{v}_1, \dots, \mathbf{v}_i, \dots, \mathbf{v}_m) + \varphi(\mathbf{v}_1, \dots, \mathbf{v}'_i, \dots, \mathbf{v}_m)$;
- (b) $\varphi(\mathbf{v}_1, \dots, a\mathbf{v}_i, \dots, \mathbf{v}_m) = a\varphi(\mathbf{v}_1, \dots, \mathbf{v}_i, \dots, \mathbf{v}_m)$.

The 2-linear maps and 3-linear maps are also called **bilinear** and **trilinear** maps, respectively.

If $U = F$ then a multilinear map into U is called a **multilinear form**.

The set of multilinear maps from $V_1 \times \cdots \times V_m$ into U , together with the operations defined as follows, is denoted $L(V_1, \dots, V_m; U)$. For m -linear maps φ, ψ , and $a \in F$,

$$\begin{aligned}(\psi + \varphi)(v_1, \dots, v_m) &= \psi(v_1, \dots, v_m) + \varphi(v_1, \dots, v_m), \\(a\varphi)(v_1, \dots, v_m) &= a\varphi(v_1, \dots, v_m).\end{aligned}$$

Let $(\mathbf{b}_{i_1}, \dots, \mathbf{b}_{i_{n_i}})$ be an ordered basis of V_i , $i = 1, \dots, m$. The set of sequences $(j_1, \dots, j_m)$, $1 \leq j_i \leq n_i$, $i = 1, \dots, m$, will be identified with the set $\Gamma(n_1, \dots, n_m)$ of maps α from $\{1, \dots, m\}$ into $\mathbb{N}$ satisfying $1 \leq \alpha(i) \leq n_i$, $i = 1, \dots, m$.

For $\alpha \in \Gamma(n_1, \dots, n_m)$, the m -tuple of basis vectors $(\mathbf{b}_{1\alpha(1)}, \dots, \mathbf{b}_{m\alpha(m)})$ is denoted by $\mathbf{b}_\alpha$.

Unless otherwise stated $\Gamma(n_1, \dots, n_m)$ is considered ordered by the lexicographic order. When there is no risk of confusion, Γ is used instead of $\Gamma(n_1, \dots, n_m)$.

Let p, q be positive integers. If φ is an $(p + q)$ -linear map from $W_1 \times \dots \times W_p \times V_1 \times \dots \times V_q$ into U , then for each choice of $\mathbf{w}_i$ in W_i , $i = 1, \dots, p$, the map

$$(\mathbf{v}_1, \dots, \mathbf{v}_q) \mapsto \varphi(\mathbf{w}_1, \dots, \mathbf{w}_p, \mathbf{v}_1, \dots, \mathbf{v}_q),$$

from $V_1 \times \dots \times V_q$ into U , is denoted $\varphi_{\mathbf{w}_1, \dots, \mathbf{w}_p}$, i.e.

$$\varphi_{\mathbf{w}_1, \dots, \mathbf{w}_p}(\mathbf{v}_1, \dots, \mathbf{v}_q) = \varphi(\mathbf{w}_1, \dots, \mathbf{w}_p, \mathbf{v}_1, \dots, \mathbf{v}_q).$$

Let η be a linear map from U into U' and θ_i a linear map from V'_i into V_i , $i = 1, \dots, m$. If $(\mathbf{v}_1, \dots, \mathbf{v}_m) \mapsto \varphi(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is a multilinear map from $V_1 \times \dots \times V_m$ into U , $L(\theta_1, \dots, \theta_m; \eta)(\varphi)$ denotes the map from $V'_1 \times \dots \times V'_m$ into U' , defined by

$$(\mathbf{v}'_1, \dots, \mathbf{v}'_m) \mapsto \eta(\varphi(\theta_1(\mathbf{v}'_1), \dots, \theta_m(\mathbf{v}'_m))).$$

Facts:

The following facts can be found in [Mar73, Chap. 1] and in [Mer97, Chap. 5].

1. If φ is a multilinear map, then $\varphi(\mathbf{v}_1, \dots, 0, \dots, \mathbf{v}_m) = 0$.
2. The set $L(V_1, \dots, V_m; U)$ is a vector space over F .
3. If φ is an m -linear map from $V_1 \times \dots \times V_m$ into U , then for every integer p , $1 \leq p < m$, and $\mathbf{v}_i \in V_i$, $1 \leq i \leq p$, the map $\varphi_{\mathbf{v}_1, \dots, \mathbf{v}_p}$ is an $(m - p)$ -linear map.
4. Under the same assumptions than in (3.) the map $(\mathbf{v}_1, \dots, \mathbf{v}_p) \mapsto \varphi_{\mathbf{v}_1, \dots, \mathbf{v}_p}$ from $V_1 \times \dots \times V_p$ into $L(V_{p+1}, \dots, V_m; U)$, is p -linear. A linear isomorphism from $L(V_1, \dots, V_p, V_{p+1}, \dots, V_m; U)$ into $L(V_1, \dots, V_p; L(V_{p+1}, \dots, V_m; U))$ arises through this construction.
5. Let η be a linear map from U into U' and θ_i a linear map from V'_i into V_i , $i = 1, \dots, m$. The map $L(\theta_1, \dots, \theta_m; \eta)$ from $L(V_1, \dots, V_m; U)$ into $L(V'_1, \dots, V'_m; U')$ is a linear map. When $m = 1$, and $U = U' = F$, then $L(\theta_1, I)$ is the dual or adjoint linear map θ_1^* from V_1^* into $V_1'^*$.
6. $|\Gamma(n_1, \dots, n_m)| = \prod_{i=1}^m n_i$ where $|\cdot|$ denotes cardinality.
7. Let $(\mathbf{y}_\alpha)_{\alpha \in \Gamma}$ be a family of vectors of U . Then, there exists a unique m -linear map φ from $V_1 \times \dots \times V_m$ into U satisfying $\varphi(\mathbf{b}_\alpha) = \mathbf{y}_\alpha$, for every $\alpha \in \Gamma$.
8. If $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ is a basis of U , then $(\varphi_{i,\alpha} : \alpha \in \Gamma, i = 1, \dots, m)$ is a basis of $L(V_1, \dots, V_m; U)$, where $\varphi_{i,\alpha}$ is characterized by the conditions $\varphi_{i,\alpha}(\mathbf{b}_\beta) = \delta_{\alpha,\beta} \mathbf{u}_i$. Moreover, if φ is an m -linear map from $V_1 \times \dots \times V_m$ into U such that for each $\alpha \in \Gamma$,

$$\varphi(\mathbf{b}_\alpha) = \sum_{i=1}^n a_{i,\alpha} \mathbf{u}_i,$$

then

$$\varphi = \sum_{\alpha, i} a_{i,\alpha} \varphi_{i,\alpha}.$$

Examples:

1. The map from F^m into F , $(a_1, \dots, a_m) \mapsto \prod_{i=1}^m a_i$, is an m -linear map.
2. Let V be a vector space over F . The map $(a, \mathbf{v}) \mapsto a\mathbf{v}$ from $F \times V$ into V is a bilinear map.
3. The map from $F^m \times F^m$ into F , $((a_1, \dots, a_m), (b_1, \dots, b_m)) \mapsto \sum a_i b_i$, is bilinear.
4. Let U, V , and W be vector spaces over F . The map $(\theta, \eta) \mapsto \theta\eta$ from $L(V, W) \times L(U, V)$ into $L(U, W)$, given by composition, is bilinear.
5. The multiplication of matrices, $(A, B) \mapsto AB$, from $F^{m \times n} \times F^{n \times p}$ into $F^{m \times p}$, is bilinear. Observe that this example is the matrix counterpart of the previous one.

6. Let V and W be vector spaces over F . The evaluation map, from $L(V, W) \times V$ into W ,

$$(\theta, \mathbf{v}) \longmapsto \theta(\mathbf{v}),$$

is bilinear.

7. The map

$$((a_{11}, a_{21}, \dots, a_{m1}), \dots, (a_{1m}, a_{2m}, \dots, a_{mm})) \rightarrow \det([a_{ij}])$$

from the Cartesian product of m copies of F^m into F is m -linear.

13.2 Tensor Products

Definitions:

Let $V_1, \dots, V_m, P$ be vector spaces over F . Let $\nu : V_1 \times \dots \times V_m \longmapsto P$ be a multilinear map. The pair (ν, P) is called a **tensor product** of $V_1, \dots, V_m$, or P is said to be a **tensor product of $V_1, \dots, V_m$ with tensor multiplication ν** , if the following condition is satisfied:

Universal factorization property

If φ is a multilinear map from $V_1 \times \dots \times V_m$ into the vector space U , then there exists a unique linear map, h , from P into U , that makes the following diagram commutative:

$$\begin{array}{ccc} V_1 \times \dots \times V_m & \xrightarrow{\nu} & P \\ & \searrow \varphi & \downarrow h \\ & & U \end{array}$$

i.e., $h\nu = \varphi$.

If P is a tensor product of $V_1, \dots, V_m$, with tensor multiplication ν , then P is denoted by $V_1 \otimes \dots \otimes V_m$ and $\nu(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is denoted by $\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_m$ and is called the **tensor product of the vectors $\mathbf{v}_1, \dots, \mathbf{v}_m$** .

The elements of $V_1 \otimes \dots \otimes V_m$ are called **tensors**. The tensors that are the tensor product of m vectors are called **decomposable tensors**.

When $V_1 = \dots = V_m = V$, the vector space $V_1 \otimes \dots \otimes V_m$ is called the m th **tensor power** of V and is denoted by $\otimes^m V$. It is convenient to define $\otimes^0 V = F$ and assume that 1 is the unique decomposable tensor of $\otimes^0 V$. When we consider simultaneously different models of tensor product, sometimes we use alternative forms to denote the tensor multiplication like $\otimes'$, $\tilde{\otimes}$, or $\hat{\otimes}$ to emphasize these different choices.

Within this section, $V_1, \dots, V_m$ are finite dimensional vector spaces over F and $(\mathbf{b}_{i1}, \dots, \mathbf{b}_{im_i})$ denotes a basis of V_i , $i = 1, \dots, m$. When V is a vector space and $\mathbf{x}_1, \dots, \mathbf{x}_k \in V$, $\text{Span}(\{\mathbf{x}_1, \dots, \mathbf{x}_k\})$ denotes the subspace of V spanned by these vectors.

Facts:

The following facts can be found in [Mar73, Chap. 1] and in [Mer97, Chap. 5].

1. If $V_1 \otimes \dots \otimes V_m$ and $V_1 \otimes' \dots \otimes' V_m$ are two tensor products of $V_1, \dots, V_m$, then the unique linear map h from $V_1 \otimes \dots \otimes V_m$ into $V_1 \otimes' \dots \otimes' V_m$ satisfying

$$h(\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_m) = \mathbf{v}_1 \otimes' \dots \otimes' \mathbf{v}_m$$

is an isomorphism.

2. If $(v(\mathbf{b}_\alpha))_{\alpha \in \Gamma(n_1, \dots, n_m)}$ is a basis of P , then the pair (v, P) is a tensor product of $V_1, \dots, V_m$. This is often the most effective way to identify a model for the tensor product of vector spaces. It also implies the existence of a tensor product.
3. If P is the tensor product of $V_1, \dots, V_m$ with tensor multiplication v , and $h : P \longrightarrow Q$ is a linear isomorphism, then (hv, Q) is a tensor product of $V_1, \dots, V_m$.
4. When $m = 1$, it makes sense to speak of a tensor product of one vector space V and V itself is used as a model for that tensor product with the identity as tensor multiplication, i.e., $\bigotimes^1 V = V$.
5. **Bilinear version of the universal property** — Given a multilinear map from $V_1 \times \cdots \times V_k \times U_1 \times \cdots \times U_m$ into W ,

$$(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{u}_1, \dots, \mathbf{u}_m) \mapsto \varphi(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{u}_1, \dots, \mathbf{u}_m),$$

there exists a unique bilinear map χ from $(V_1 \otimes \cdots \otimes V_k) \times (U_1 \otimes \cdots \otimes U_m)$ into W satisfying

$$\begin{aligned} \chi(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_k, \mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_m) &= \varphi(\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{u}_1, \dots, \mathbf{u}_m), \\ \mathbf{v}_i &\in V_i \quad \mathbf{u}_j \in U_j, \quad i = 1, \dots, k, \quad j = 1, \dots, m. \end{aligned}$$

6. Let $a \in F$ and $\mathbf{v}_i, \mathbf{v}'_i \in V_i$, $i = 1, \dots, m$. As the consequence of the multilinearity of $\otimes$, the following equalities hold:
 - (a) $\mathbf{v}_1 \otimes \cdots \otimes (\mathbf{v}_i + \mathbf{v}'_i) \otimes \cdots \otimes \mathbf{v}_m$
 $= \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_i \otimes \cdots \otimes \mathbf{v}_m + \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}'_i \otimes \cdots \otimes \mathbf{v}_m,$
 - (b) $a(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_m) = (a\mathbf{v}_1) \otimes \cdots \otimes \mathbf{v}_m = \cdots = \mathbf{v}_1 \otimes \cdots \otimes (a\mathbf{v}_m),$
 - (c) $\mathbf{v}_1 \otimes \cdots \otimes 0 \otimes \cdots \otimes \mathbf{v}_m = 0.$
7. If one of the vector spaces V_i is zero, then $V_1 \otimes \cdots \otimes V_m = \{0\}$.
8. Write $\mathbf{b}_\alpha^\otimes$ to mean

$$\mathbf{b}_\alpha^\otimes := \mathbf{b}_{1\alpha(1)} \otimes \cdots \otimes \mathbf{b}_{m\alpha(m)}.$$

Then

$$(\mathbf{b}_\alpha^\otimes)_{\alpha \in \Gamma}$$

is a basis of $V_1 \otimes \cdots \otimes V_m$. This basis is said to be induced by the bases $(\mathbf{b}_{i1}, \dots, \mathbf{b}_{im_i})$, $i = 1, \dots, m$.

9. The decomposable tensors span the tensor product $V_1 \otimes \cdots \otimes V_m$. Furthermore, if the set C_i spans V_i , $i = 1, \dots, m$, then the set $\{\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_m : \mathbf{v}_i \in C_i, i = 1, \dots, m\}$ spans $V_1 \otimes \cdots \otimes V_m$.
10. $\dim(V_1 \otimes \cdots \otimes V_m) = \prod_{i=1}^m \dim(V_i)$.
11. The tensor product is commutative,

$$V_1 \otimes V_2 = V_2 \otimes V_1,$$

meaning that if $V_1 \otimes V_2$ is a tensor product of V_1 and V_2 , then $V_1 \otimes V_2$ is also a tensor product of V_2 and V_1 with tensor multiplication $(\mathbf{v}_2, \mathbf{v}_1) \mapsto \mathbf{v}_1 \otimes \mathbf{v}_2$.

In general, with a similar meaning, for any $\sigma \in S_m$,

$$V_1 \otimes \cdots \otimes V_m = V_{\sigma(1)} \otimes \cdots \otimes V_{\sigma(m)}.$$

12. The tensor product is associative,

$$(V_1 \otimes V_2) \otimes V_3 = V_1 \otimes (V_2 \otimes V_3) = V_1 \otimes V_2 \otimes V_3,$$

meaning that:

- (a) A tensor product $V_1 \otimes V_2 \otimes V_3$ is also a tensor product of $V_1 \otimes V_2$ and V_3 (respectively of V_1 and $V_2 \otimes V_3$) with tensor multiplication defined (uniquely by Fact 5 above) for $\mathbf{v}_i \in V_i, i = 1, 2, 3$, by $(\mathbf{v}_1 \otimes \mathbf{v}_2) \otimes \mathbf{v}_3 = \mathbf{v}_1 \otimes \mathbf{v}_2 \otimes \mathbf{v}_3$ (respectively by $\mathbf{v}_1 \otimes (\mathbf{v}_2 \otimes \mathbf{v}_3) = \mathbf{v}_1 \otimes \mathbf{v}_2 \otimes \mathbf{v}_3$).
- (b) And, $V_1 \otimes (V_2 \otimes V_3)$ (respectively $V_1 \otimes (V_2 \otimes V_3)$) is a tensor product of V_1, V_2, V_3 with tensor multiplication defined by $\mathbf{v}_1 \otimes \mathbf{v}_2 \otimes \mathbf{v}_3 = (\mathbf{v}_1 \otimes \mathbf{v}_2) \otimes \mathbf{v}_3, \mathbf{v}_i \in V_i, i = 1, 2, 3$ (respectively $\mathbf{v}_1 \otimes \mathbf{v}_2 \otimes \mathbf{v}_3 = \mathbf{v}_1 \otimes (\mathbf{v}_2 \otimes \mathbf{v}_3), \mathbf{v}_i \in V_i, i = 1, 2, 3$).

In general, with an analogous meaning,

$$(V_1 \otimes \cdots \otimes V_k) \otimes (V_{k+1} \otimes \cdots \otimes V_m) = V_1 \otimes \cdots \otimes V_m,$$

for any $k, 1 \leq k < m$.

13. Let W_i be a subspace of $V_i, i = 1, \dots, m$. Then $W_1 \otimes \cdots \otimes W_m$ is a subspace of $V_1 \otimes \cdots \otimes V_m$, meaning that the subspace of $V_1 \otimes \cdots \otimes V_m$ spanned by the set of decomposable tensors of the form

$$\mathbf{w}_1 \otimes \cdots \otimes \mathbf{w}_m, \quad \mathbf{w}_i \in W_i, \quad i = 1, \dots, m$$

is a tensor product of $W_1, \dots, W_m$ with tensor multiplication equal to the restriction of $\otimes$ to $W_1 \times \cdots \times W_m$.

From now on, the model for the tensor product described above is assumed when dealing with the tensor product of subspaces of V_i .

14. Let W_1, W'_1 be subspaces of V_1 and W_2 and W'_2 be subspaces of V_2 . Then

$$(a) \quad (W_1 \otimes W_2) \cap (W'_1 \otimes W'_2) = (W_1 \cap W'_1) \otimes (W_2 \cap W'_2).$$

$$(b) \quad W_1 \otimes (W_2 + W'_2) = (W_1 \otimes W_2) + (W_1 \otimes W'_2),$$

$$(W_1 + W'_1) \otimes W_2 = (W_1 \otimes W_2) + (W'_1 \otimes W_2).$$

$$(c) \quad \text{Assuming } W_1 \cap W'_1 = \{0\},$$

$$(W_1 \oplus W'_1) \otimes W_2 = (W_1 \otimes W_2) \oplus (W'_1 \otimes W_2).$$

$$\text{Assuming } W_2 \cap W'_2 = \{0\},$$

$$W_1 \otimes (W_2 \oplus W'_2) = (W_1 \otimes W_2) \oplus (W_1 \otimes W'_2).$$

15. In a more general setting, if $W_{ij}, j = 1, \dots, p_i$ are subspaces of $V_i, i \in \{1, \dots, m\}$, then

$$\left(\sum_{j=1}^{p_1} W_{1j} \right) \otimes \cdots \otimes \left(\sum_{j=1}^{p_m} W_{mj} \right) = \sum_{\gamma \in \Gamma(p_1, \dots, p_m)} W_{1\gamma(1)} \otimes \cdots \otimes W_{m\gamma(m)}.$$

If the sums of subspaces in the left-hand side are direct, then

$$\left(\bigoplus_{j=1}^{p_1} W_{1j} \right) \otimes \cdots \otimes \left(\bigoplus_{j=1}^{p_m} W_{mj} \right) = \bigoplus_{\gamma \in \Gamma(p_1, \dots, p_m)} W_{1\gamma(1)} \otimes \cdots \otimes W_{m\gamma(m)}.$$

Examples:

1. The vector space $F^{m \times n}$ of the $m \times n$ matrices over F is a tensor product of F^m and F^n with tensor multiplication (the usual tensor multiplication for $F^{m \times n}$) defined, for $(a_1, \dots, a_m) \in F^m$ and $(b_1, \dots, b_n) \in F^n$, by

$$(a_1, \dots, a_m) \otimes (b_1, \dots, b_n) = \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix} \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix}.$$

With this definition, $\mathbf{e}_i \otimes \mathbf{e}'_j = E_{ij}$ where $\mathbf{e}_i$, $\mathbf{e}'_j$, and E_{ij} are standard basis vectors of F^m , F^n , and $F^{m \times n}$.

2. The field F , viewed as a vector space over F , is an m th tensor power of F with tensor multiplication defined by

$$a_1 \otimes \cdots \otimes a_m = \prod_{i=1}^m a_i, \quad a_i \in F, \quad i = 1, \dots, m.$$

3. The vector space V is a tensor product of F and V with tensor multiplication defined by

$$a \otimes \mathbf{v} = a\mathbf{v}, \quad a \in F, \quad \mathbf{v} \in V.$$

4. Let U and V be vector spaces over F . Then $L(V; U)$ is a tensor product $U \otimes V^*$ with tensor multiplication (the usual tensor multiplication for $L(V; U)$) defined by the equality $(\mathbf{u} \otimes f)(\mathbf{v}) = f(\mathbf{v})\mathbf{u}$, $\mathbf{u} \in U$, $\mathbf{v} \in V$.
5. Let $V_1, \dots, V_m$ be vector spaces over F . The vector space $L(V_1, \dots, V_m; U)$ is a tensor product $L(V_1, \dots, V_m; F) \otimes U$ with tensor multiplication

$$(\varphi \otimes \mathbf{u})(\mathbf{v}_1, \dots, \mathbf{v}_m) = \varphi(\mathbf{v}_1, \dots, \mathbf{v}_m)\mathbf{u}.$$

6. Denote by $F^{n_1 \times \cdots \times n_m}$ the set of all families with elements indexed in $\{1, \dots, n_1\} \times \cdots \times \{1, \dots, n_m\} = \Gamma(n_1, \dots, n_m)$. The set $F^{n_1 \times \cdots \times n_m}$ equipped with the sum and scalar product defined, for every $(j_1, \dots, j_m) \in \Gamma(n_1, \dots, n_m)$, by the equalities

$$(a_{j_1, \dots, j_m}) + (b_{j_1, \dots, j_m}) = (a_{j_1, \dots, j_m} + b_{j_1, \dots, j_m}),$$

$$\alpha(a_{j_1, \dots, j_m}) = (\alpha a_{j_1, \dots, j_m}), \quad \alpha \in F,$$

is a vector space over F . This vector space is a tensor product of $F^{n_1}, \dots, F^{n_m}$ with tensor multiplication defined by

$$(a_{11}, \dots, a_{1n_1}) \otimes \cdots \otimes (a_{m1}, \dots, a_{mn_m}) = \left(\prod_{i=1}^m a_{ij_i} \right)_{(j_1, \dots, j_m) \in \Gamma}.$$

7. The vector space $L(V_1, \dots, V_m; F)$ is a tensor product of $V_1^* = L(V_1; F), \dots, V_m^* = L(V_m; F)$ with tensor multiplication defined by

$$g_1 \otimes \cdots \otimes g_m(\mathbf{v}_1, \dots, \mathbf{v}_m) = \prod_{t=1}^m g_t(\mathbf{v}_t).$$

Very often, for example in the context of geometry, the factors of the tensor product are vector space duals. In those situations, this is the model of tensor product implicitly assumed.

8. The vector space

$$L(V_1, \dots, V_m; F)^*$$

is a tensor product of $V_1, \dots, V_m$ with tensor multiplication defined by

$$\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_m(\psi) = \psi(\mathbf{v}_1, \dots, \mathbf{v}_m).$$

9. The vector space $L(V_1, \dots, V_m; F)$ is a tensor product $L(V_1, \dots, V_k; F) \otimes L(V_{k+1}, \dots, V_m; F)$ with tensor multiplication defined, for every $\mathbf{v}_i \in V_i$, $i = 1, \dots, m$, by the equalities

$$(\varphi \otimes \psi)(\mathbf{v}_1, \dots, \mathbf{v}_m) = \varphi(\mathbf{v}_1, \dots, \mathbf{v}_k) \psi(\mathbf{v}_{k+1}, \dots, \mathbf{v}_m).$$

13.3 Rank of a Tensor: Decomposable Tensors

Definitions:

Let $z \in V_1 \otimes \cdots \otimes V_m$. The tensor z has **rank** k if z is the sum of k decomposable tensors but it cannot be written as sum of l decomposable tensors, for any l less than k .

Facts:

The following facts can be found in [Bou89, Chap. II, §7.8] and [Mar73, Chap. 1].

1. The tensor $z = \mathbf{v}_1 \otimes \mathbf{w}_1 + \cdots + \mathbf{v}_t \otimes \mathbf{w}_t \in V \otimes W$ has rank t if and only if $(\mathbf{v}_1, \dots, \mathbf{v}_t)$ and $(\mathbf{w}_1, \dots, \mathbf{w}_t)$ are linearly independent.
2. If the model for the tensor product of F^m and F^n is the vector space of $m \times n$ matrices over F with the usual tensor multiplication, then the rank of a tensor is equal to the rank of the corresponding matrix.
3. If the model for the tensor product $U \otimes V^*$ is the vector space $L(V; U)$ with the usual tensor multiplication, then the rank of a tensor is equal to the rank of the corresponding linear map.
4. $\mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_m = 0$ if and only if $\mathbf{x}_i = 0$ for some $i \in \{1, \dots, m\}$.
5. If $\mathbf{x}_i, \mathbf{y}_i$ are nonzero vectors of V_i , $i = 1, \dots, m$, then

$$\text{Span}(\{\mathbf{x}_1 \otimes \cdots \otimes \mathbf{x}_m\}) = \text{Span}(\{\mathbf{y}_1 \otimes \cdots \otimes \mathbf{y}_m\})$$

if and only if $\text{Span}(\{\mathbf{x}_i\}) = \text{Span}(\{\mathbf{y}_i\})$, $i = 1, \dots, m$.

Examples:

1. Consider as a model of $F^m \otimes F^n$, the vector space of the $m \times n$ matrices over F with the usual tensor multiplication. Let A be a tensor of $F^m \otimes F^n$. If $\text{rank } A = k$ (using the matrix definition of rank), then

$$A = M \begin{bmatrix} I_k & 0 \\ 0 & 0 \end{bmatrix} N,$$

where $M = [\mathbf{x}_1 \cdots \mathbf{x}_m]$ is an invertible matrix with columns $\mathbf{x}_1, \dots, \mathbf{x}_m$ and

$$N = \begin{bmatrix} \mathbf{y}_1 \\ \mathbf{y}_2 \\ \vdots \\ \mathbf{y}_n \end{bmatrix}$$

is an invertible matrix with rows $\mathbf{y}_1, \dots, \mathbf{y}_n$. (See Chapter 2.) Then

$$A = \mathbf{x}_1 \otimes \mathbf{y}_1 + \cdots + \mathbf{x}_k \otimes \mathbf{y}_k$$

has rank k as a tensor.

13.4 Tensor Product of Linear Maps

Definitions:

Let θ_i be a linear map from V_i into U_i , $i = 1, \dots, m$. The unique linear map h from $V_1 \otimes \dots \otimes V_m$ into $U_1 \otimes \dots \otimes U_m$ satisfying, for all $\mathbf{v}_i \in V_i$, $i = 1, \dots, m$,

$$h(\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_m) = \theta_1(\mathbf{v}_1) \otimes \dots \otimes \theta_m(\mathbf{v}_m)$$

is called the **tensor product** of $\theta_1, \dots, \theta_m$ and is denoted by $\theta_1 \otimes \dots \otimes \theta_m$.

Let $A_t = (a_{ij}^{(t)})$ be an $r_t \times s_t$ matrix over F , $t = 1, \dots, m$. The **Kronecker product** of $A_1, \dots, A_m$, denoted $A_1 \otimes \dots \otimes A_m$, is the $(\prod_{t=1}^m r_t) \times (\prod_{t=1}^m s_t)$ matrix whose (α, β) -entry ($\alpha \in \Gamma(r_1, \dots, r_m)$ and $\beta \in \Gamma(s_1, \dots, s_m)$) is $\prod_{t=1}^m a_{\alpha(t)\beta(t)}^{(t)}$. (See also Section 10.4.)

Facts:

The following facts can be found in [Mar73, Chap. 2] and in [Mer97, Chap. 5].

Let θ_i be a linear map from V_i into U_i , $i = 1, \dots, m$.

1. If η_i is a linear map from W_i into V_i , $i = 1, \dots, m$,

$$(\theta_1 \otimes \dots \otimes \theta_m)(\eta_1 \otimes \dots \otimes \eta_m) = (\theta_1 \eta_1) \otimes \dots \otimes (\theta_m \eta_m).$$

2. $I_{V_1 \otimes \dots \otimes V_m} = I_{V_1} \otimes \dots \otimes I_{V_m}$.

3. $\text{Ker}(\theta_1 \otimes \dots \otimes \theta_m) = \text{Ker}(\theta_1) \otimes V_2 \otimes \dots \otimes V_m + V_1 \otimes \text{Ker}(\theta_2) \otimes \dots \otimes V_m + \dots + V_1 \otimes \dots \otimes V_{m-1} \otimes \text{Ker}(\theta_m)$.

In particular, $\theta_1 \otimes \dots \otimes \theta_m$ is one to one if θ_i is one to one, $i = 1, \dots, m$, [Bou89, Chap. II, §3.5].

4. $\theta_1 \otimes \dots \otimes \theta_m(V_1 \otimes \dots \otimes V_m) = \theta_1(V_1) \otimes \dots \otimes \theta_m(V_m)$. In particular $\theta_1 \otimes \dots \otimes \theta_m$ is onto if θ_i is onto, $i = 1, \dots, m$.

In the next three facts, assume that θ_i is a linear operator on the n_i -dimensional vector space V_i , $i = 1, \dots, m$.

5. $\text{tr}(\theta_1 \otimes \dots \otimes \theta_m) = \prod_{i=1}^m \text{tr}(\theta_i)$.

6. If $\sigma(\theta_i) = \{a_{i1}, \dots, a_{in_i}\}$, $i = 1, \dots, m$, then

$$\sigma(\theta_1 \otimes \dots \otimes \theta_m) = \left\{ \prod_{i=1}^m a_{i, \alpha(i)} \right\}_{\alpha \in \Gamma(n_1, \dots, n_m)}.$$

7. $\det(\theta_1 \otimes \theta_2 \otimes \dots \otimes \theta_m) = \det(\theta_1)^{n_2 \dots n_m} \det(\theta_2)^{n_1 \dots n_m} \dots \det(\theta_m)^{n_1 \dots n_{m-1}}$.

8. The map $\nu : (\theta_1, \dots, \theta_m) \mapsto \theta_1 \otimes \dots \otimes \theta_m$ is a multilinear map from $L(V_1; U_1) \times \dots \times L(V_m; U_m)$ into $L(V_1 \otimes \dots \otimes V_m; U_1 \otimes \dots \otimes U_m)$.

9. The vector space $L(V_1 \otimes \dots \otimes V_m; U_1 \otimes \dots \otimes U_m)$ is a tensor product of the vector spaces $L(V_1; U_1), \dots, L(V_m; U_m)$, with tensor multiplication $(\theta_1, \dots, \theta_m) \mapsto \theta_1 \otimes \dots \otimes \theta_m$:

$$L(V_1; U_1) \otimes \dots \otimes L(V_m; U_m) = L(V_1 \otimes \dots \otimes V_m; U_1 \otimes \dots \otimes U_m).$$

10. As a consequence of (9.), choosing F as the model for $\bigotimes^m F$ with the product in F as tensor multiplication,

$$V_1^* \otimes \dots \otimes V_m^* = (V_1 \otimes \dots \otimes V_m)^*.$$

11. Let $(\mathbf{v}_{ij})_{j=1,\dots,n_i}$ be an ordered basis of V_i and $(\mathbf{u}_{ij})_{j=1,\dots,q_i}$ an ordered basis of U_i , $i = 1, \dots, m$. Let A_i be the matrix of θ_i on the bases fixed in V_i and U_i . Then the matrix of $\theta_1 \otimes \dots \otimes \theta_m$ on the bases $(\mathbf{v}_\alpha^\otimes)_{\alpha \in \Gamma(n_1, \dots, n_m)}$ and $(\mathbf{u}_\alpha^\otimes)_{\alpha \in \Gamma(q_1, \dots, q_m)}$ (induced by the bases $(\mathbf{v}_{ij})_{j=1,\dots,n_i}$ and $(\mathbf{u}_{ij})_{j=1,\dots,q_i}$, respectively) is the Kronecker product of $A_1, \dots, A_m$,

$$A_1 \otimes \dots \otimes A_m.$$

12. Let $n_1, \dots, n_m, r_1, \dots, r_m, t_1, \dots, t_m$ be positive integers. Let A_i be an $n_i \times r_i$ matrix, and B_i be an $r_i \times t_i$ matrix, $i = 1, \dots, m$. Then the following holds:

- (a) $(A_1 \otimes \dots \otimes A_m)(B_1 \otimes \dots \otimes B_m) = A_1 B_1 \otimes \dots \otimes A_m B_m$,
 (b) $(A_1 \otimes \dots \otimes A_k) \otimes (A_{k+1} \otimes \dots \otimes A_m) = A_1 \otimes \dots \otimes A_m$.

Examples:

1. Consider as a model of $U \otimes V^*$, the vector space $L(V; U)$ with tensor multiplication defined by $(\mathbf{u} \otimes f)(\mathbf{v}) = f(\mathbf{v})\mathbf{u}$. Use a similar model for the tensor product of U' and V'^* . Let $\eta \in L(U; U')$ and $\theta \in L(V'; V)$. Then, for all $\xi \in U \otimes V^* = L(V; U)$,

$$\eta \otimes \theta^*(\xi) = \eta \xi \theta.$$

2. Consider as a model of $F^m \otimes F^n$, the vector space of the $m \times n$ matrices over F with the usual tensor multiplication. Use a similar model for the tensor product of F^r and F^s . Identify the set of column matrices, $F^{m \times 1}$, with F^m and the set of row matrices, $F^{1 \times n}$, with F^n . Let A be an $r \times m$ matrix over F . Let θ_A be the linear map from F^m into F^r defined by

$$\theta_A(a_1, \dots, a_m) = A \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{bmatrix}.$$

Let B be an $s \times n$ matrix. Then, for all $C \in F^{m \times n} = F^m \otimes F^n$, $\theta_A \otimes \theta_B(C) = ACB^T$.

3. For every $i = 1, \dots, m$ consider the ordered basis $(\mathbf{b}_{i1}, \dots, \mathbf{b}_{in_i})$ fixed in V_i , and the basis $(\mathbf{b}'_{i1}, \dots, \mathbf{b}'_{is_i})$ fixed in U_i . Let θ_i be a linear map from V_i into U_i and let $A_i = (a_{jk}^{(i)})$ be the $s_i \times n_i$ matrix of θ_i with respect to the bases $(\mathbf{b}_{i1}, \dots, \mathbf{b}_{in_i})$, $(\mathbf{b}'_{i1}, \dots, \mathbf{b}'_{is_i})$. For every $z \in V_1 \otimes \dots \otimes V_m$,

$$\begin{aligned} z &= \sum_{j_1=1}^{n_1} \sum_{j_2=1}^{n_2} \dots \sum_{j_m=1}^{n_m} c_{j_1, \dots, j_m} \mathbf{b}_{1j_1} \otimes \dots \otimes \mathbf{b}_{mj_m} \\ &= \sum_{\alpha \in \Gamma(n_1, \dots, n_m)} c_\alpha \mathbf{b}_\alpha^\otimes. \end{aligned}$$

Then, for $\beta = (i_1, \dots, i_m) \in \Gamma(s_1, \dots, s_m)$, the component $c'_{i_1, \dots, i_m}$ of $\theta_1 \otimes \dots \otimes \theta_m(z)$ on the basis element $\mathbf{b}'_{1i_1} \otimes \dots \otimes \mathbf{b}'_{mi_m}$ of $U_1 \otimes \dots \otimes U_m$ is

$$\begin{aligned} c'_\beta &= c'_{i_1, \dots, i_m} = \sum_{j_1=1}^{n_1} \dots \sum_{j_m=1}^{n_m} a_{i_1, j_1}^{(1)} \dots a_{i_m, j_m}^{(m)} c_{j_1, \dots, j_m} \\ &= \sum_{\gamma \in \Gamma(n_1, \dots, n_m)} \left(\prod_{i=1}^m a_{\beta(i)\gamma(i)}^{(i)} \right) c_\gamma. \end{aligned}$$

4. If $A = [a_{ij}]$ is an $p \times q$ matrix over F and B is an $r \times s$ matrix over F , then the Kronecker product of A and B is the matrix whose partition in $r \times s$ blocks is

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B & \cdots & a_{1q}B \\ a_{21}B & a_{22}B & \cdots & a_{2q}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{p1}B & a_{p2}B & \cdots & a_{pq}B \end{bmatrix}.$$

13.5 Symmetric and Antisymmetric Maps

Recall that we are assuming F to be of characteristic zero and that all vector spaces are finite dimensional over F . In particular, V and U denote finite dimensional vector spaces over F .

Definitions:

Let m be a positive integer. When $V_1 = V_2 = \cdots = V_m = V$ $L^m(V; U)$ denotes the vector space of the multilinear maps $L(V_1, \dots, V_m; U)$. By convention $L^0(V; U) = U$.

An m -linear map $\psi \in L^m(V; U)$ is called **antisymmetric or alternating** if it satisfies

$$\psi(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(m)}) = \text{sgn}(\sigma) \psi(\mathbf{v}_1, \dots, \mathbf{v}_m), \quad \sigma \in S_m,$$

where $\text{sgn}(\sigma)$ denotes the sign of the permutation σ .

Similarly, an m -linear map $\varphi \in L^m(V; U)$ satisfying

$$\varphi(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(m)}) = \varphi(\mathbf{v}_1, \dots, \mathbf{v}_m)$$

for all permutations $\sigma \in S_m$ and for all $\mathbf{v}_1, \dots, \mathbf{v}_m$ in V is called **symmetric**. Let $S^m(V; U)$ and $A^m(V; U)$ denote the subsets of $L^m(V; U)$ whose elements are respectively the symmetric and the antisymmetric m -linear maps. The elements of $A^m(V; F)$ are called **antisymmetric forms**. The elements of $S^m(V; F)$ are called **symmetric forms**.

Let $\Gamma_{m,n}$ be the set of all maps from $\{1, \dots, m\}$ into $\{1, \dots, n\}$, i.e.,

$$\Gamma_{m,n} = \Gamma(\underbrace{n, \dots, n}_{m \text{ times}}).$$

The subset of $\Gamma_{m,n}$ of the strictly increasing maps α ($\alpha(1) < \cdots < \alpha(m)$) is denoted by $Q_{m,n}$. The subset of the increasing maps $\alpha \in \Gamma_{m,n}$ ($\alpha(1) \leq \cdots \leq \alpha(m)$) is denoted by $G_{m,n}$.

Let $A = [a_{ij}]$ be an $m \times n$ matrix over F . Let $\alpha \in \Gamma_{p,m}$ and $\beta \in \Gamma_{q,n}$. Then $A[\alpha|\beta]$ be the $p \times q$ -matrix over F whose (i, j) -entry is $a_{\alpha(i), \beta(j)}$, i.e.,

$$A[\alpha|\beta] = [a_{\alpha(i), \beta(j)}].$$

The m th-tuple $(1, 2, \dots, m)$ is denoted by ι_m . If there is no risk of confusion ι is used instead of ι_m .

Facts:

1. If $m > n$, we have $Q_{m,n} = \emptyset$. The cardinality of $\Gamma_{m,n}$ is n^m , the cardinality of $Q_{m,n}$ is $\binom{n}{m}$, and the cardinality of $G_{m,n}$ is $\binom{m+n-1}{m}$.
2. $A^m(V; U)$ and $S^m(V; U)$ are vector subspaces of $L^m(V; U)$.
3. Let $\psi \in L^m(V; U)$. The following conditions are equivalent:

- (a) ψ is an antisymmetric multilinear map.
- (b) For $1 \leq i < j \leq m$ and for all $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$,

$$\begin{aligned} \psi(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_j, \mathbf{v}_{i+1}, \dots, \mathbf{v}_{j-1}, \mathbf{v}_i, \mathbf{v}_{j+1}, \dots, \mathbf{v}_m) \\ = -\psi(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots, \mathbf{v}_{j-1}, \mathbf{v}_j, \mathbf{v}_{j+1}, \dots, \mathbf{v}_m). \end{aligned}$$

- (c) For $1 \leq i < m$ and for all $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$,

$$\psi(\mathbf{v}_1, \dots, \mathbf{v}_{i+1}, \mathbf{v}_i, \dots, \mathbf{v}_m) = -\psi(\mathbf{v}_1, \dots, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots, \mathbf{v}_m).$$

4. Let $\psi \in L^m(V; U)$. The following conditions are equivalent:

- (a) ψ is a symmetric multilinear map.
- (b) For $1 \leq i < j \leq m$ and for all $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$,

$$\begin{aligned} \psi(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_j, \mathbf{v}_{i+1}, \dots, \mathbf{v}_{j-1}, \mathbf{v}_i, \mathbf{v}_{j+1}, \dots, \mathbf{v}_m) \\ = \psi(\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots, \mathbf{v}_{j-1}, \mathbf{v}_j, \mathbf{v}_{j+1}, \dots, \mathbf{v}_m). \end{aligned}$$

- (c) For $1 \leq i < m$ and for all $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$,

$$\psi(\mathbf{v}_1, \dots, \mathbf{v}_{i+1}, \mathbf{v}_i, \dots, \mathbf{v}_m) = \psi(\mathbf{v}_1, \dots, \mathbf{v}_i, \mathbf{v}_{i+1}, \dots, \mathbf{v}_m).$$

5. When we consider $L^m(V; U)$ as the tensor product, $L^m(V; F) \otimes U$, with the tensor multiplication described in Example 5 in Section 13.2, we have

$$A^m(V; U) = A^m(V; F) \otimes U \quad \text{and} \quad S^m(V; U) = S^m(V; F) \otimes U.$$

6. **Polarization identity** [Dol04] If φ is a symmetric multilinear map, then for every m -tuple $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ of vectors of V , and for any vector $\mathbf{w} \in V$, the following identity holds:

$$\begin{aligned} \varphi(\mathbf{v}_1, \dots, \mathbf{v}_m) &= \\ &= \frac{1}{2^m m!} \sum_{\varepsilon_1 \dots \varepsilon_m} \varepsilon_1 \dots \varepsilon_m \varphi(\mathbf{w} + \varepsilon_1 \mathbf{v}_1 + \dots + \varepsilon_m \mathbf{v}_m, \dots, \mathbf{w} + \varepsilon_1 \mathbf{v}_1 + \dots + \varepsilon_m \mathbf{v}_m), \end{aligned}$$

where $\varepsilon_i \in \{-1, +1\}$, $i = 1, \dots, m$.

Examples:

1. The map

$$((a_{11}, a_{21}, \dots, a_{m1}), \dots, (a_{1m}, a_{2m}, \dots, a_{mm})) \rightarrow \det([a_{ij}])$$

from the Cartesian product of m copies of F^m into F is m -linear and antisymmetric.

2. The map

$$((a_{11}, a_{21}, \dots, a_{m1}), \dots, (a_{1m}, a_{2m}, \dots, a_{mm})) \rightarrow \text{per}([a_{ij}])$$

from the Cartesian product of m copies of F^m into F is m -linear and symmetric.

3. The map $((a_1, \dots, a_n), (b_1, \dots, b_n)) \mapsto (a_i b_j - b_i a_j)$ from $F^n \times F^n$ into $F^{n \times n}$ is bilinear anti-symmetric.
4. The map $((a_1, \dots, a_n), (b_1, \dots, b_n)) \mapsto (a_i b_j + b_i a_j)$ from $F^n \times F^n$ into $F^{n \times n}$ is bilinear symmetric.
5. The map χ from V^m into $A^m(V; F)^*$ defined by

$$\chi(\mathbf{v}_1, \dots, \mathbf{v}_m)(\psi) = \psi(\mathbf{v}_1, \dots, \mathbf{v}_m), \quad \mathbf{v}_1, \dots, \mathbf{v}_m \in V,$$

is an antisymmetric multilinear map.

6. The map χ from V^m into $S^m(V; F)^*$ defined by

$$\chi(\mathbf{v}_1, \dots, \mathbf{v}_m)(\psi) = \psi(\mathbf{v}_1, \dots, \mathbf{v}_m), \quad \mathbf{v}_1, \dots, \mathbf{v}_m \in V,$$

is a symmetric multilinear map.

13.6 Symmetric and Grassmann Tensors

Definitions:

Let $\sigma \in S_m$ be a permutation of $\{1, \dots, m\}$. The unique linear map, from $\otimes^m V$ into $\otimes^m V$ satisfying

$$\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_m \mapsto \mathbf{v}_{\sigma^{-1}(1)} \otimes \dots \otimes \mathbf{v}_{\sigma^{-1}(m)}, \quad \mathbf{v}_1, \dots, \mathbf{v}_m \in V,$$

is denoted $P(\sigma)$.

Let ψ be a multilinear form of $L^m(V; F)$ and σ an element of S_m . The multilinear form $(\mathbf{v}_1, \dots, \mathbf{v}_m) \mapsto \psi(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(m)})$ is denoted ψ_σ .

The linear operator Alt from $\otimes^m V$ into $\otimes^m V$ defined by

$$\text{Alt} := \frac{1}{m!} \sum_{\sigma \in S_m} \text{sgn}(\sigma) P(\sigma)$$

is called the **alternator**. In order to emphasize the degree of the domain of Alt , Alt_m is often used for the operator having $\otimes^m V$, as domain.

Similarly, the linear operator Sym is defined as the following linear combination of the maps $P(\sigma)$:

$$\text{Sym} = \frac{1}{m!} \sum_{\sigma \in S_m} P(\sigma).$$

As before, Sym_m is often written to mean the Sym operator having $\otimes^m V$, as domain.

The range of Alt is denoted by $\bigwedge^m V$, i.e., $\bigwedge^m V = \text{Alt}(\otimes^m V)$, and is called the **Grassmann space of degree m** associated with V or the **m th-exterior power** of V .

The range of Sym is denoted by $\bigvee^m V$, i.e., $\bigvee^m V = \text{Sym}(\otimes^m V)$, and is called the **symmetric space of degree m** associated with V or the **m th symmetric power** of V .

By convention

$$\bigotimes^0 V = \bigwedge^0 V = \bigvee^0 V = F.$$

Assume $m \geq 1$. The elements of $\bigwedge^m V$ that are the image under Alt of decomposable tensors of $\bigotimes^m V$ are called **decomposable elements of $\bigwedge^m V$** . If $\mathbf{x}_1, \dots, \mathbf{x}_m \in V$, $\mathbf{x}_1 \wedge \dots \wedge \mathbf{x}_m$ denotes the decomposable element of $\bigwedge^m V$,

$$\mathbf{x}_1 \wedge \dots \wedge \mathbf{x}_m = m! \text{Alt}(\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_m),$$

and $\mathbf{x}_1 \wedge \dots \wedge \mathbf{x}_m$ is called the **exterior product** of $\mathbf{x}_1, \dots, \mathbf{x}_m$. Similarly, the elements of $\bigvee^m V$ that are the image under Sym of decomposable tensors of $\bigotimes^m V$ are called **decomposable elements of $\bigvee^m V$** . If $\mathbf{x}_1, \dots, \mathbf{x}_m \in V$, $\mathbf{x}_1 \vee \dots \vee \mathbf{x}_m$ denotes the decomposable element of $\bigvee^m V$,

$$\mathbf{x}_1 \vee \dots \vee \mathbf{x}_m = m! \text{Sym}(\mathbf{x}_1 \otimes \dots \otimes \mathbf{x}_m),$$

and $\mathbf{x}_1 \vee \dots \vee \mathbf{x}_m$ is called the **symmetric product** of $\mathbf{x}_1, \dots, \mathbf{x}_m$.

Let $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ be a basis of V . If $\alpha \in \Gamma_{m,n}$, $\mathbf{b}_\alpha^\otimes$, $\mathbf{b}_\alpha^\wedge$, and $\mathbf{b}_\alpha^\vee$ denote respectively the tensors

$$\mathbf{b}_\alpha^\otimes = \mathbf{b}_{\alpha(1)} \otimes \dots \otimes \mathbf{b}_{\alpha(m)},$$

$$\mathbf{b}_\alpha^\wedge = \mathbf{b}_{\alpha(1)} \wedge \dots \wedge \mathbf{b}_{\alpha(m)},$$

$$\mathbf{b}_\alpha^\vee = \mathbf{b}_{\alpha(1)} \vee \dots \vee \mathbf{b}_{\alpha(m)}.$$

Let n and m be positive integers. An n -**composition** of m is a sequence

$$\mu = (\mu_1, \dots, \mu_n)$$

of nonnegative integers that sum to m . Let $\mathcal{C}_{m,n}$ be the set of n -compositions of m .

Let $\lambda = (\lambda_1, \dots, \lambda_n)$ be an n -composition of m . The integer $\lambda_1! \dots \lambda_n!$ will be denoted by $\lambda!$.

Let $\alpha \in \Gamma_{m,n}$. The **multiplicity composition** of α is the n -tuple of the cardinalities of the fibers of α , $(|\alpha^{-1}(1)|, \dots, |\alpha^{-1}(n)|)$, and is denoted by λ_α .

Facts:

The following facts can be found in [Mar73, Chap. 2], [Mer97, Chap. 5], and [Spi79, Chap. 7].

1. $\bigwedge^m V$ and $\bigvee^m V$ are vector subspaces of $\bigotimes^m V$.
2. The map $\sigma \mapsto P(\sigma)$ from the symmetric group of degree m into $L(\bigotimes^m V; \bigotimes^m V)$ is an F -representation of S_m , i.e., $P(\sigma\tau) = P(\sigma)P(\tau)$ for any $\sigma, \tau \in S_m$ and $P(I) = I_{\bigotimes^m V}$.
3. Choosing $L^m(V; F)$, with the usual tensor multiplication, as the model for the tensor power, $\bigotimes^m V^*$, the linear operator $P(\sigma)$ acts on $L^m(V; F)$ by the following transformation

$$(P(\sigma)\psi) = \psi_\sigma.$$

4. The linear operators Alt and Sym are projections, i.e., $\text{Alt}^2 = \text{Alt}$ and $\text{Sym}^2 = \text{Sym}$.
5. If $m = 1$, we have

$$\text{Sym} = \text{Alt} = I_{\bigotimes^1 V} = I_V.$$

6. $\bigwedge^m V = \{z \in \bigotimes^m V : P(\sigma)(z) = \text{sgn}(\sigma)z, \forall \sigma \in S_m\}$.
7. $\bigvee^m V = \{z \in \bigotimes^m V : P(\sigma)(z) = z, \forall \sigma \in S_m\}$.
8. Choosing $L^m(V; F)$ as the model for the tensor power $\bigotimes^m V^*$ with the usual tensor multiplication,

$$\bigwedge^m V^* = A^m(V; F) \quad \text{and} \quad \bigvee^m V^* = S^m(V; F).$$

9.

$$\bigotimes^1 V = \bigwedge^1 V = \bigvee^1 V = V.$$

10. $\bigotimes^2 V = \bigwedge^2 V \oplus \bigvee^2 V$. Moreover for $z \in \bigotimes^2 V$,

$$z = \text{Alt}(z) + \text{Sym}(z).$$

The corresponding equality is no more true in $\bigotimes^m V$ if $m \neq 2$.

11. $\bigwedge^m V = \{0\}$ if $m > \dim(V)$.
12. If $m \geq 1$, any element of $\bigwedge^m V$ is a sum of decomposable elements of $\bigwedge^m V$.
13. If $m \geq 1$, any element of $\bigvee^m V$ is a sum of decomposable elements of $\bigvee^m V$.
14. $\text{Alt}(P(\sigma)z) = \text{sgn}(\sigma)\text{Alt}(z)$ and $\text{Sym}(P(\sigma)(z)) = \text{Sym}(z)$, $z \in \bigotimes^m V$.
15. The map $\wedge$ from V^m into $\bigwedge^m V$ defined for $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$ by

$$\wedge(\mathbf{v}_1, \dots, \mathbf{v}_m) = \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_m$$

is an antisymmetric m -linear map.

16. The map $\vee$ from V^m into $\bigvee^m V$ defined for $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$ by

$$\vee(\mathbf{v}_1, \dots, \mathbf{v}_m) = \mathbf{v}_1 \vee \cdots \vee \mathbf{v}_m$$

is a symmetric m -linear map.

17. **(Universal property for $\bigwedge^m V$)** Given an antisymmetric m -linear map ψ from V^m into U , there exists a unique linear map h from $\bigwedge^m V$ into U such that

$$\psi(\mathbf{v}_1, \dots, \mathbf{v}_m) = h(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_m), \quad \mathbf{v}_1, \dots, \mathbf{v}_m \in V,$$

i.e., there exists a unique linear map h that makes the following diagram commutative:

$$\begin{array}{ccc} V^m & \xrightarrow{\wedge} & \bigwedge^m V \\ & \searrow \psi & \downarrow h \\ & & U \end{array}$$

18. **(Universal property for $\bigvee^m V$)** Given a symmetric m -linear map φ from V^m into U , there exists a unique linear map h from $\bigvee^m V$ into U such that

$$\varphi(\mathbf{v}_1, \dots, \mathbf{v}_m) = h(\mathbf{v}_1 \vee \cdots \vee \mathbf{v}_m), \quad \mathbf{v}_1, \dots, \mathbf{v}_m \in V,$$

i.e., there exists a unique linear map h that makes the following diagram commutative:

$$\begin{array}{ccc} V^m & \xrightarrow{\vee} & \bigvee^m V \\ & \searrow \varphi & \downarrow h \\ & & U \end{array}$$

Let p and q be positive integers.

19. **(Universal property for $\otimes^m V$ -bilinear version)** If ψ is a $(p+q)$ -linear map from V^{p+q} into U , then there exists a unique bilinear map χ from $\otimes^p V \times \otimes^q V$ into U satisfying (recall Fact 5 in Section 13.2)

$$\chi(\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_p, \mathbf{v}_{p+1} \otimes \cdots \otimes \mathbf{v}_{p+q}) = \psi(\mathbf{v}_1, \dots, \mathbf{v}_{p+q}).$$

20. **(Universal property for $\wedge^m V$ -bilinear version)** If ψ is a $(p+q)$ -linear map from V^{p+q} into U antisymmetric in the first p variables and antisymmetric in the last q variables, then there exists a unique bilinear map χ from $\wedge^p V \times \wedge^q V$ into U satisfying

$$\chi(\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_p, \mathbf{v}_{p+1} \wedge \cdots \wedge \mathbf{v}_{p+q}) = \psi(\mathbf{v}_1, \dots, \mathbf{v}_{p+q}).$$

21. **(Universal property for $\vee^m V$ -bilinear version)** If φ is a $(p+q)$ -linear map from V^{p+q} into U symmetric in the first p variables and symmetric in the last q variables, then there exists a unique bilinear map χ from $\vee^p V \times \vee^q V$ into U satisfying

$$\chi(\mathbf{v}_1 \vee \cdots \vee \mathbf{v}_p, \mathbf{v}_{p+1} \vee \cdots \vee \mathbf{v}_{p+q}) = \varphi(\mathbf{v}_1, \dots, \mathbf{v}_{p+q}).$$

22. If $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ is a basis of V , then $(\mathbf{b}_\alpha^\otimes)_{\alpha \in \Gamma_{m,n}}$ is a basis of $\otimes^m V$, $(\mathbf{b}_\alpha^\wedge)_{\alpha \in Q_{m,n}}$ is a basis of $\wedge^m V$, and $(\mathbf{b}_\alpha^\vee)_{\alpha \in G_{m,n}}$ is a basis of $\vee^m V$. These bases are said to be **induced** by the basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$.
 23. Assume $L^m(V; F)$ as the model for the tensor power of $\otimes^m V^*$, with the usual tensor multiplication. Let $(f_1, \dots, f_n)$ be the dual basis of the basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$. Then:

- (a) For every $\varphi \in L^m(V; F)$,

$$\varphi = \sum_{\alpha \in \Gamma_{m,n}} \varphi(\mathbf{b}_\alpha) f_\alpha^\otimes.$$

- (b) For every $\varphi \in A^m(V, F)$,

$$\varphi = \sum_{\alpha \in Q_{m,n}} \varphi(\mathbf{b}_\alpha) f_\alpha^\wedge.$$

- (c) For every $\varphi \in S^m(V, F)$,

$$\varphi = \sum_{\alpha \in G_{m,n}} \frac{1}{\lambda_\alpha!} \varphi(\mathbf{b}_\alpha) f_\alpha^\vee.$$

24. $\dim \otimes^m V = n^m$, $\dim \wedge^m V = \binom{n}{m}$, and $\dim \vee^m V = \binom{n+m-1}{m}$.

25. The family

$$((\mu_1 \mathbf{b}_1 + \cdots + \mu_n \mathbf{b}_n) \vee \cdots \vee (\mu_1 \mathbf{b}_1 + \cdots + \mu_n \mathbf{b}_n))_{\mu \in \mathcal{C}_{m,n}}$$

is a basis of $\vee^m V$ [Mar73, Chap. 3].

26. Let $\mathbf{x}_1, \dots, \mathbf{x}_m$ be vectors of V and $g_1, \dots, g_m$ forms of V^* . Let $a_{ij} = g_i(\mathbf{x}_j)$, $i, j = 1, \dots, m$. Then, choosing $(\otimes^m V)^*$ as the model for $\otimes^m V^*$ with tensor multiplication as described in Fact 10 in Section 13.4,

$$g_1 \otimes \cdots \otimes g_m(\mathbf{x}_1 \wedge \cdots \wedge \mathbf{x}_m) = \det[a_{ij}].$$

27. Under the same conditions of the former fact,

$$g_1 \otimes \cdots \otimes g_m(\mathbf{x}_1 \vee \cdots \vee \mathbf{x}_m) = \text{per}[a_{ij}].$$

28. Let $(f_1, \dots, f_n)$ be the dual basis of the basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$. Then, choosing $(\otimes^m V)^*$ as the model for $\otimes^m V^*$:

(a)

$$(f_\alpha^\otimes)_{\alpha \in \Gamma_{m,n}}$$

is the dual basis of the basis $(\mathbf{b}_\alpha^\otimes)_{\alpha \in \Gamma_{m,n}}$ of $\otimes^m V$.

(b)

$$\left((f_\alpha^\otimes)|_{\bigwedge^m V} \right)_{\alpha \in Q_{m,n}}$$

is the dual basis of the basis $(\mathbf{b}_\alpha^\wedge)_{\alpha \in Q_{m,n}}$ of $\bigwedge^m V$.

(c)

$$\left(\frac{1}{\lambda_\alpha!} (f_\alpha^\otimes)|_{\bigvee^m V} \right)_{\alpha \in G_{m,n}}$$

is the dual basis of the basis $(\mathbf{b}_\alpha^\vee)_{\alpha \in G_{m,n}}$ of $\bigvee^m V$.

Let $\mathbf{v}_1, \dots, \mathbf{v}_m$ be vectors of V and $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ be a basis of V .

29. Let $A = [a_{ij}]$ be the $n \times m$ matrix over F such that $\mathbf{v}_j = \sum_{i=1}^n a_{ij} \mathbf{b}_i$, $j = 1, \dots, m$. Then:

(a)

$$\mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_m = \sum_{\alpha \in \Gamma_{m,n}} \left(\prod_{t=1}^m a_{\alpha(t),t} \right) \mathbf{b}_\alpha^\otimes;$$

(b)

$$\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_m = \sum_{\alpha \in Q_{m,n}} \det A[\alpha|\iota] \mathbf{b}_\alpha^\wedge;$$

(c)

$$\mathbf{v}_1 \vee \cdots \vee \mathbf{v}_m = \sum_{\alpha \in G_{m,n}} \frac{1}{\lambda_\alpha!} \text{per} A[\alpha|\iota] \mathbf{b}_\alpha^\vee.$$

30. $\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_m = 0$ if and only if $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ is linearly dependent.

31. $\mathbf{v}_1 \vee \cdots \vee \mathbf{v}_m = 0$ if and only if one of the $\mathbf{v}_i$ s is equal to 0.

32. Let $\mathbf{u}_1, \dots, \mathbf{u}_m$ be vectors of V .

(a) If $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ are linearly independent families, then

$$\text{Span}(\{\mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_m\}) = \text{Span}(\{\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_m\})$$

if and only if

$$\text{Span}(\{\mathbf{u}_1, \dots, \mathbf{u}_m\}) = \text{Span}(\{\mathbf{v}_1, \dots, \mathbf{v}_m\}).$$

(b) If $(\mathbf{v}_1, \dots, \mathbf{v}_m)$ and $(\mathbf{u}_1, \dots, \mathbf{u}_m)$ are families of nonzero vectors of V , then

$$\text{Span}(\{\mathbf{v}_1 \vee \cdots \vee \mathbf{v}_m\}) = \text{Span}(\{\mathbf{u}_1 \vee \cdots \vee \mathbf{u}_m\})$$

if and only if there exists a permutation σ of S_m satisfying

$$\text{Span}(\{\mathbf{v}_i\}) = \text{Span}(\{\mathbf{u}_{\sigma(i)}\}), \quad i = 1, \dots, m.$$

Examples:

1. If $m = 1$, we have

$$\text{Sym} = \text{Alt} = I_{\otimes^1 V} = I_V.$$

2. Consider as a model of $\otimes^2 F^n$, the vector space of the $n \times n$ matrices with the usual tensor multiplication. Then $\wedge^2 F^n$ is the subspace of the $n \times n$ antisymmetric matrices over F and $\vee^2 F^n$ is the subspace of the $n \times n$ symmetric matrices over F . Moreover, for $(a_1, \dots, a_n), (b_1, \dots, b_n) \in F^n$:

$$(a) \quad (a_1, \dots, a_n) \wedge (b_1, \dots, b_n) = [a_i b_j - b_i a_j]_{i,j=1,\dots,n}.$$

$$(b) \quad (a_1, \dots, a_n) \vee (b_1, \dots, b_n) = [a_i b_j + b_i a_j]_{i,j=1,\dots,n}.$$

With these definitions, $\mathbf{e}_i \wedge \mathbf{e}_j = E_{ij} - E_{ji}$ and $\mathbf{e}_i \vee \mathbf{e}_j = E_{ij} + E_{ji}$, where $\mathbf{e}_i, \mathbf{e}_j$, and E_{ij} are standard basis vectors of F^m, F^n , and $F^{m \times n}$.

3. For $\mathbf{x} \in V$, $\mathbf{x} \vee \dots \vee \mathbf{x} = m! \mathbf{x} \otimes \dots \otimes \mathbf{x}$.

13.7 The Tensor Multiplication, the Alt Multiplication, and the Sym Multiplication

Next we will introduce “external multiplications” for tensor powers, Grassmann spaces, and symmetric spaces. Let p, q be positive integers.

Definitions:

The (p, q) -**tensor multiplication** is the unique bilinear map, $(z, z') \mapsto z \otimes z'$ from $(\otimes^p V) \times (\otimes^q V)$ into $\otimes^{p+q} V$, satisfying

$$(\mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_p) \otimes (\mathbf{v}_{p+1} \otimes \dots \otimes \mathbf{v}_{p+q}) = \mathbf{v}_1 \otimes \dots \otimes \mathbf{v}_{p+q}.$$

The (p, q) -**alt multiplication** (briefly alt multiplication) is the unique bilinear map (recall Fact 20 in section 13.6), $(z, z') \mapsto z \wedge z'$ from $(\wedge^p V) \times (\wedge^q V)$ into $\wedge^{p+q} V$, satisfying

$$(\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_p) \wedge (\mathbf{v}_{p+1} \wedge \dots \wedge \mathbf{v}_{p+q}) = \mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_{p+q}.$$

The (p, q) -**sym multiplication** (briefly sym multiplication) is the unique bilinear map (recall Fact 21 in section 13.6), $(z, z') \mapsto z \vee z'$ from $(\vee^p V) \times (\vee^q V)$ into $\vee^{p+q} V$, satisfying

$$(\mathbf{v}_1 \vee \dots \vee \mathbf{v}_p) \vee (\mathbf{v}_{p+1} \vee \dots \vee \mathbf{v}_{p+q}) = \mathbf{v}_1 \vee \dots \vee \mathbf{v}_{p+q}.$$

These definitions can be extended to include the cases where either p or q is zero, taking as multiplication the scalar product.

Let m, n be positive integers satisfying $1 \leq m < n$. Let $\alpha \in Q_{m,n}$. We denote by α^c the element of $Q_{n-m,n}$ whose range is the complement in $\{1, \dots, n\}$ of the range of α and by $\tilde{\alpha}$ the permutation of S_n :

$$\tilde{\alpha} = \begin{pmatrix} 1 & \dots & m & m+1 & \dots & n \\ \alpha(1) & \dots & \alpha(m) & \alpha^c(1) & \dots & \alpha^c(n) \end{pmatrix}.$$

Facts:

The following facts can be found in [Mar73, Chap. 2], [Mer97, Chap. 5], and in [Spi79, Chap. 7].

1. The value of the alt multiplication for arbitrary elements $z \in \bigwedge^p V$ and $z' \in \bigwedge^q V$ is given by

$$z \wedge z' = \frac{(p+q)!}{p!q!} \text{Alt}_{p+q}(z \otimes z').$$

2. The product of $z \in \bigvee^p V$ and $z' \in \bigvee^q V$ by the sym multiplication is given by

$$z \vee z' = \frac{(p+q)!}{p!q!} \text{Sym}_{p+q}(z \otimes z').$$

3. The alt-multiplication $z \wedge z'$ and the sym-multiplication $z \vee z'$ are not, in general, decomposable elements of any Grassmann or symmetric space of degree 2.
4. Let $0 \neq z \in \bigwedge^m V$. Then z is decomposable if and only if there exists a linearly independent family of vectors $\mathbf{v}_1, \dots, \mathbf{v}_m$ satisfying $z \wedge \mathbf{v}_i = 0, i = 1, \dots, m$.
5. If $\dim(V) = n$, all elements of $\bigwedge^{n-1} V$ are decomposable.
6. The multiplications defined in this subsection are associative. Therefore,

$$\begin{aligned} z \otimes z' \otimes z'', z &\in \bigotimes^p V, \quad z' \in \bigotimes^q V, \quad z'' \in \bigotimes^r V; \\ w \wedge w' \wedge w'', w &\in \bigwedge^p V, \quad w' \in \bigwedge^q V, \quad w'' \in \bigwedge^r V; \\ y \vee y' \vee y'', y &\in \bigvee^p V, \quad y' \in \bigvee^q V, \quad y'' \in \bigvee^r V \end{aligned}$$

are meaningful as well as similar expressions with more than three factors.

7. If $w \in \bigwedge^p V, w' \in \bigwedge^q V$, then

$$w' \wedge w = (-1)^{pq} w \wedge w'.$$

8. If $y \in \bigvee^p V, y' \in \bigvee^q V$, then

$$y' \vee y = y \vee y'.$$

Examples:

1. When the vector space is the dual $V^* = L(V; F)$ of a vector space and we choose as the models of tensor powers of V^* the spaces of multilinear forms (with the usual tensor multiplication), then the image of the tensor multiplication $\varphi \otimes \psi$ ($\varphi \in L^p(V; F)$ and $\psi \in L^q(V; F)$) on $(\mathbf{v}_1, \dots, \mathbf{v}_{p+q})$ is given by the equality

$$(\varphi \otimes \psi)(\mathbf{v}_1, \dots, \mathbf{v}_{p+q}) = \varphi(\mathbf{v}_1, \dots, \mathbf{v}_p) \psi(\mathbf{v}_{p+1}, \dots, \mathbf{v}_{p+q}).$$

2. When the vector space is the dual $V^* = L(V; F)$ of a vector space and we choose as the models for the tensor powers of V^* the spaces of multilinear forms (with the usual tensor multiplication), the alt multiplication of $\varphi \in A^p(V; F)$ and $\psi \in A^q(V; F)$ takes the form

$$\begin{aligned} &(\varphi \wedge \psi)(\mathbf{v}_1, \dots, \mathbf{v}_{p+q}) \\ &= \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \text{sgn}(\sigma) \varphi(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(p)}) \psi(\mathbf{v}_{\sigma(p+1)}, \dots, \mathbf{v}_{\sigma(p+q)}). \end{aligned}$$

3. The equality in Example 2 has an alternative expression that can be seen as a “Laplace expansion” for antisymmetric forms

$$\begin{aligned}
& (\varphi \wedge \psi)(\mathbf{v}_1, \dots, \mathbf{v}_{p+q}) \\
&= \sum_{\alpha \in Q_{p,p+q}} \text{sgn}(\tilde{\alpha}) \varphi(\mathbf{v}_{\alpha(1)}, \dots, \mathbf{v}_{\alpha(p)}) \psi(\mathbf{v}_{\alpha^c(1)}, \dots, \mathbf{v}_{\alpha^c(q)}).
\end{aligned}$$

4. In the case $p = 1$, the equality in Example 3 has the form

$$\begin{aligned}
& (\varphi \wedge \psi)(\mathbf{v}_1, \dots, \mathbf{v}_{q+1}) \\
&= \sum_{j=1}^{q+1} (-1)^{j+1} \varphi(\mathbf{v}_j) \psi(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_{q+1}).
\end{aligned}$$

5. When the vector space is the dual $V^* = L(V; F)$ of a vector space and we choose as the models of tensor powers of V^* the spaces of multilinear forms (with the usual tensor multiplication), the value of sym multiplication of $\varphi \in S^p(V; F)$ and $\psi \in S^q(V; F)$ on $(\mathbf{v}_1, \dots, \mathbf{v}_{p+q})$ is

$$\begin{aligned}
& (\varphi \vee \psi)(\mathbf{v}_1, \dots, \mathbf{v}_{p+q}) \\
&= \frac{1}{p!q!} \sum_{\sigma \in S_{p+q}} \varphi(\mathbf{v}_{\sigma(1)}, \dots, \mathbf{v}_{\sigma(p)}) \psi(\mathbf{v}_{\sigma(p+1)}, \dots, \mathbf{v}_{\sigma(p+q)}).
\end{aligned}$$

6. The equality in Example 5 has an alternative expression that can be seen as a “Laplace expansion” for symmetric forms

$$\begin{aligned}
& (\varphi \vee \psi)(\mathbf{v}_1, \dots, \mathbf{v}_{p+q}) \\
&= \sum_{\alpha \in Q_{p,p+q}} \varphi(\mathbf{v}_{\alpha(1)}, \dots, \mathbf{v}_{\alpha(p)}) \psi(\mathbf{v}_{\alpha^c(1)}, \dots, \mathbf{v}_{\alpha^c(q)}).
\end{aligned}$$

7. In the case $p = 1$, the equality in Example 6 has the form

$$\begin{aligned}
& (\varphi \vee \psi)(\mathbf{v}_1, \dots, \mathbf{v}_{q+1}) \\
&= \sum_{j=1}^{q+1} \varphi(\mathbf{v}_j) \psi(\mathbf{v}_1, \dots, \mathbf{v}_{j-1}, \mathbf{v}_{j+1}, \dots, \mathbf{v}_{q+1}).
\end{aligned}$$

13.8 Associated Maps

Definitions:

Let $\theta \in L(V; U)$. The linear map $\theta \otimes \dots \otimes \theta$ from $\bigotimes^m V$ into $\bigotimes^m U$ (the tensor product of m copies of θ) will be denoted by $\bigotimes^m \theta$. The subspaces $\bigwedge^m V$ and $\bigvee^m V$ are mapped by $\bigotimes^m \theta$ into $\bigwedge^m U$ and $\bigvee^m U$, respectively. The restriction of $\bigotimes^m \theta$ to $\bigwedge^m V$ and to $\bigvee^m V$ will be respectively denoted, $\bigwedge^m \theta$ e $\bigvee^m \theta$.

Facts:

The following facts can be found in [Mar73, Chap. 2].

1. Let $\mathbf{v}_1, \dots, \mathbf{v}_m \in V$. The following properties hold:
 - (a) $\bigwedge^m \theta(\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_m) = \theta(\mathbf{v}_1) \wedge \dots \wedge \theta(\mathbf{v}_m)$.
 - (b) $\bigvee^m \theta(\mathbf{v}_1 \vee \dots \vee \mathbf{v}_m) = \theta(\mathbf{v}_1) \vee \dots \vee \theta(\mathbf{v}_m)$.
2. Let $\theta \in L(V; U)$ and $\eta \in L(W, V)$. The following equalities hold:
 - (a) $\bigwedge^m(\theta\eta) = \bigwedge^m(\theta) \bigwedge^m(\eta)$.
 - (b) $\bigvee^m(\theta\eta) = \bigvee^m(\theta) \bigvee^m(\eta)$.
3. $\bigwedge^m(I_V) = I_{\bigwedge^m V}$; $\bigvee^m(I_V) = I_{\bigvee^m V}$.
4. Let $\theta, \eta \in L(V; U)$ and assume that $\text{rank}(\theta) > m$. Then

$$\bigwedge^m \theta = \bigwedge^m \eta$$

if and only if $\theta = a\eta$ and $a^m = 1$.

5. Let $\theta, \eta \in L(V; U)$. Then $\bigvee^m \theta = \bigvee^m \eta$ if and only if $\theta = a\eta$ and $a^m = 1$.
6. If θ is one-to-one (respectively onto), then $\bigwedge^m \theta$ and $\bigvee^m \theta$ are one-to-one (respectively onto).
7. From now on θ is a linear operator on the n -dimensional vector space V .
7. Considering $\bigwedge^n \theta$ as an operator in the one-dimensional space $\bigwedge^n V$,

$$\left(\bigwedge^n \theta \right)(z) = \det(\theta)z, \text{ for all } z \in \bigwedge^n V.$$

8. If the characteristic polynomial of θ is

$$\mathbf{p}_\theta(x) = x^n + \sum_{i=1}^n (-1)^i a_i x^{n-i},$$

then

$$a_i = \text{tr} \left(\bigwedge^i \theta \right), \quad i = 1, \dots, n.$$

9. If θ has spectrum $\sigma(\theta) = \{\lambda_1, \dots, \lambda_n\}$, then

$$\sigma \left(\bigwedge^m \theta \right) = \left\{ \prod_{i=1}^m \lambda_{\alpha(i)} \right\}_{\alpha \in Q_{m,n}}, \quad \sigma \left(\bigvee^m \theta \right) = \left\{ \prod_{i=1}^m \lambda_{\alpha(i)} \right\}_{\alpha \in G_{m,n}}.$$

10.

$$\det \left(\bigwedge^m \theta \right) = \det(\theta)^{\binom{n-1}{m-1}}, \quad \det \left(\bigvee^m \theta \right) = \det(\theta)^{\binom{m+n-1}{m-1}}.$$

Examples:

1. Let A be the matrix of the linear operator $\theta \in L(V; V)$ in the basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$. The linear operator on $\bigwedge^m V$ whose matrix in the basis $(\mathbf{b}_\alpha^\wedge)_{\alpha \in Q_{m,n}}$ is the m th compound of A is $\bigwedge^m \theta$.

13.9 Tensor Algebras

Definitions:

Let A be an F -algebra and $(A_k)_{k \in \mathbb{N}}$ a family of vector subspaces of A . The algebra A is **graded** by $(A_k)_{k \in \mathbb{N}}$ if the following conditions are satisfied:

- (a) $A = \bigoplus_{k \in \mathbb{N}} A_k$.
- (b) $A_i A_j \subseteq A_{i+j}$ for every $i, j \in \mathbb{N}$.

The elements of A_k are known as **homogeneous of degree k** , and the elements of $\bigcup_{n \in \mathbb{N}} A_k$ are called **homogeneous**.

By condition (a), every element of A can be written uniquely as a sum of (a finite number of nonzero) homogeneous elements, i.e., given $u \in A$ there exist uniquely determined $\mathbf{u}_k \in A_k, k \in \mathbb{N}$ satisfying

$$\mathbf{u} = \sum_{k \in \mathbb{N}} \mathbf{u}_k.$$

These elements are called **homogeneous components of $\mathbf{u}$** . The summand of degree k in the former equation is denoted by $[\mathbf{u}]_k$.

From now on V is a finite dimensional vector space over F of dimension n . As before $\bigotimes^k V$ denotes the k th-tensor power of V .

Denote by $\bigotimes V$ the external direct sum of the vector spaces $\bigotimes^k V, k \in \mathbb{N}$. If $z_i \in \bigotimes^i V, z_i$ is identified with the sequence $z \in \bigotimes V$ whose i th coordinate is z_i and the remaining coordinates are 0. Therefore, after this identification,

$$\bigotimes V = \bigoplus_{k \in \mathbb{N}} \bigotimes^k V.$$

Consider in $\bigotimes V$ the multiplication $(x, y) \mapsto x \otimes y$ defined for $x, y \in \bigotimes V$ by

$$[x \otimes y]_k = \sum_{\substack{r, s \in \mathbb{N} \\ r+s=k}} [x]_r \otimes [y]_s, \quad k \in \mathbb{N},$$

where $[x]_r \otimes [y]_s$ is the (r, s) -tensor multiplication of $[x]_r$ and $[y]_s$ introduced in the definitions of Section 13.7. The vector space $\bigotimes V$ equipped with this multiplication is called the **tensor algebra** on V .

Denote by $\bigwedge V$ the external direct sum of the vector spaces $\bigwedge^k V, k \in \mathbb{N}$. If $z_i \in \bigwedge^i V, z_i$ is identified with the sequence $z \in \bigwedge V$ whose i th coordinate is z_i and the remaining coordinates are 0. Therefore, after this identification,

$$\bigwedge V = \bigoplus_{k \in \mathbb{N}} \bigwedge^k V.$$

Recall that $\bigwedge^k V = \{0\}$ if $k > n$. Then

$$\bigwedge V = \bigoplus_{k=0}^n \bigwedge^k V$$

and the elements of $\bigwedge V$ can be uniquely written in the form

$$z_0 + z_1 + \cdots + z_n, \quad z_i \in \bigwedge^i V, \quad i = 0, \dots, n.$$

Consider in $\bigwedge V$ the multiplication $(x, y) \mapsto x \wedge y$ defined, for $x, y \in \bigwedge V$, by

$$[x \wedge y]_k = \sum_{\substack{r, s \in \{0, \dots, n\} \\ r+s=k}} [x]_r \wedge [y]_s, \quad k \in \{0, \dots, n\},$$

where $[x]_r \wedge [y]_s$ is the (r, s) -alt multiplication of $[x]_r$ and $[y]_s$ referred in definitions of Section 13.7. The vector space $\bigwedge V$ equipped with this multiplication is called the **Grassmann algebra** on V .

Denote by $\bigvee V$ the external direct sum of the vector spaces $\bigvee^k V, k \in \mathbb{N}$.

If $z_i \in \bigvee^i V$, we identify z_i with the sequence $z \in \bigvee V$ whose i th coordinate is z_i and the remaining coordinates are 0. Therefore, after this identification

$$\bigvee V = \bigoplus_{k \in \mathbb{N}} \bigvee^k V.$$

Consider in $\bigvee V$ the multiplication $(x, y) \mapsto x \vee y$ defined for $x, y \in \bigvee V$ by

$$[x \vee y]_k = \sum_{\substack{r, s \in \mathbb{N} \\ r+s=k}} [x]_r \vee [y]_s, \quad k \in \mathbb{N},$$

where $[x]_r \vee [y]_s$ is the (r, s) -sym multiplication of $[x]_r$ and $[y]_s$ referred in definitions of Section 13.7.

The vector space $\bigvee V$ equipped with this multiplication is called the **symmetric algebra** on V .

Facts:

The following facts can be found in [Mar73, Chap. 3] and [Gre67, Chaps. II and III].

1. The vector space $\bigotimes V$ with the multiplication $(x, y) \mapsto x \otimes y$ is an algebra over F graded by $(\bigotimes^k V)_{k \in \mathbb{N}}$, whose identity is the identity of $F = \bigotimes^0 V$.
2. The vector space $\bigwedge V$ with the multiplication $(x, y) \mapsto x \wedge y$ is an algebra over F graded by $(\bigwedge^k V)_{k \in \mathbb{N}}$ whose identity is the identity of $F = \bigwedge^0 V$.
3. The vector space $\bigvee V$ with the multiplication $(x, y) \mapsto x \vee y$ is an algebra over F graded by $(\bigvee^k V)_{k \in \mathbb{N}}$ whose identity is the identity of $F = \bigvee^0 V$.
4. The F -algebra $\bigotimes V$ does not have zero divisors.
5. Let B be an F -algebra and θ a linear map from V into B satisfying $\theta(\mathbf{x})\theta(\mathbf{y}) = -\theta(\mathbf{y})\theta(\mathbf{x})$ for all $\mathbf{x}, \mathbf{y} \in V$. Then there exists a unique algebra homomorphism h from $\bigwedge V$ into B satisfying $h|_V = \theta$.
6. Let B be an F -algebra and θ a linear map from V into B satisfying $\theta(\mathbf{x})\theta(\mathbf{y}) = \theta(\mathbf{y})\theta(\mathbf{x})$, for all $\mathbf{x}, \mathbf{y} \in V$. Then there exists a unique algebra homomorphism h from $\bigvee V$ into B satisfying $h|_V = \theta$.
7. Let $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ be a basis of V . The symmetric algebra $\bigvee^m V$ is isomorphic to the algebra of polynomials in n indeterminates, $F[x_1, \dots, x_n]$, by the algebra isomorphism whose restriction to V is the linear map that maps $\mathbf{b}_i$ into x_i , $i = 1, \dots, n$.

Examples:

1. Let $x_1, \dots, x_n$ be n distinct indeterminates. Let V be the vector space of the formal linear combinations with coefficients in F in the indeterminates $x_1, \dots, x_n$. The tensor algebra on V is the algebra of the polynomials in the noncommuting indeterminates $x_1, \dots, x_n$ ([Coh03], [Jac64]). This algebra is denoted by

$$F \langle x_1, \dots, x_n \rangle.$$

The elements of this algebra are of the form

$$f(x_1, \dots, x_n) = \sum_{m \in \mathbb{N}} \sum_{\alpha \in \Gamma_{m,n}} c_\alpha x_{\alpha(1)} \otimes \cdots \otimes x_{\alpha(m)},$$

with all but a finite number of the coefficients c_α equal to zero.

13.10 Tensor Product of Inner Product Spaces

Unless otherwise stated, within this section V, U , and W , as well as these letters subscripts, superscripts, or accents, are finite dimensional vector spaces over $\mathbb{R}$ or over $\mathbb{C}$, equipped with an inner product.

The inner product of V is denoted by $\langle \cdot, \cdot \rangle_V$. When there is no risk of confusion $\langle \cdot, \cdot \rangle$ is used instead. In this section F means either the field $\mathbb{R}$ or the field $\mathbb{C}$.

Definitions:

Let θ be a linear map from V into W . The notation θ^* will be used for the **adjoint** of θ (i.e., the linear map from W into V satisfying $\langle \theta(\mathbf{x}), \mathbf{y} \rangle = \langle \mathbf{x}, \theta^*(\mathbf{y}) \rangle$ for all $\mathbf{x} \in V$ and $\mathbf{y} \in W$).

The unique inner product $\langle \cdot, \cdot \rangle$ on $V_1 \otimes \cdots \otimes V_m$ satisfying, for every $\mathbf{v}_i, \mathbf{u}_i \in V_i$, $i = 1, \dots, m$,

$$\langle \mathbf{v}_1 \otimes \cdots \otimes \mathbf{v}_m, \mathbf{u}_1 \otimes \cdots \otimes \mathbf{u}_m \rangle = \prod_{i=1}^m \langle \mathbf{v}_i, \mathbf{u}_i \rangle_{V_i},$$

is called **induced inner product** associated with the inner products $\langle \cdot, \cdot \rangle_{V_i}$, $i = 1, \dots, m$.

For each $\mathbf{v} \in V$, let $f_{\mathbf{v}} \in V^*$ be defined by $f_{\mathbf{v}}(\mathbf{u}) = \langle \mathbf{u}, \mathbf{v} \rangle$. The inverse of the map $\mathbf{v} \rightarrow f_{\mathbf{v}}$ is denoted by ϱ_V (briefly ϱ). The inner product on V^* , defined by

$$\langle f, g \rangle = \langle \varrho(g), \varrho(f) \rangle_V,$$

is called the **dual** of $\langle \cdot, \cdot \rangle_V$.

Let U, V be inner product spaces over F . We consider defined in $L(V; U)$ the **Hilbert–Schmidt** inner product, i.e., the inner product defined, for $\theta, \eta \in L(V; U)$, by $\langle \theta, \eta \rangle = \text{tr}(\eta^* \theta)$.

From now on $V_1 \otimes \cdots \otimes V_m$ is assumed to be equipped with the inner product induced by the inner products $\langle \cdot, \cdot \rangle_{V_i}$, $i = 1, \dots, m$.

Facts:

The following facts can be found in [Mar73, Chap. 2].

1. The map $\mathbf{v} \rightarrow f_{\mathbf{v}}$ is bijective-linear if $F = \mathbb{R}$ and conjugate-linear (i.e., $c\mathbf{v} \mapsto \bar{c}f_{\mathbf{v}}$) if $F = \mathbb{C}$.
2. If $(\mathbf{b}_{i1}, \dots, \mathbf{b}_{in_i})$ is an orthonormal basis of V_i , $i = 1, \dots, m$, then $\{\mathbf{b}_{\alpha}^{\otimes} : \alpha \in \Gamma(n_1, \dots, n_m)\}$ is an orthonormal basis of $V_1 \otimes \cdots \otimes V_m$.
3. Let $\theta_i \in L(V_i; W_i)$, $i = 1, \dots, m$, with adjoint map $\theta_i^* \in L(W_i; V_i)$. Then,

$$(\theta_1 \otimes \cdots \otimes \theta_m)^* = \theta_1^* \otimes \cdots \otimes \theta_m^*.$$

4. If $\theta_i \in L(V_i; V_i)$ is Hermitian (normal, unitary), $i = 1, \dots, m$, then $\theta_1 \otimes \cdots \otimes \theta_m$ is also Hermitian (normal, unitary).
5. Let $\theta \in L(V; V)$. If $\bigotimes^m \theta$ ($\bigvee^m \theta$) is normal, then θ is normal.
6. Let $\theta \in L(V; V)$. Assume that θ is a linear operator on V with rank greater than m . If $\bigwedge^m \theta$ is normal, then θ is normal.
7. If $\mathbf{u}_1, \dots, \mathbf{u}_m, \mathbf{v}_1, \dots, \mathbf{v}_m \in V$:

$$\langle \mathbf{u}_1 \wedge \cdots \wedge \mathbf{u}_m, \mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_m \rangle = m! \det \langle \mathbf{u}_i, \mathbf{v}_j \rangle,$$

$$\langle \mathbf{u}_1 \vee \cdots \vee \mathbf{u}_m, \mathbf{v}_1 \vee \cdots \vee \mathbf{v}_m \rangle = m! \text{per} \langle \mathbf{u}_i, \mathbf{v}_j \rangle.$$

8. Let $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ be an orthonormal basis of V . Then the basis $(\mathbf{b}_{\alpha}^{\otimes})_{\alpha \in \Gamma_{m,n}}$ is an orthonormal basis of $\bigotimes^m V$, $(\sqrt{\frac{1}{m!}} \mathbf{b}_{\alpha}^{\wedge})_{\alpha \in Q_{m,n}}$ is an orthonormal basis of $\bigwedge^m V$, and $(\sqrt{\frac{1}{m! \lambda_{\alpha}!}} \mathbf{b}_{\alpha}^{\vee})_{\alpha \in G_{m,n}}$ is an orthonormal basis of $\bigvee^m V$.

Examples:

The field F (recall that $F = \mathbb{R}$ or $F = \mathbb{C}$) has an inner product, $(a, b) \mapsto \langle a, b \rangle = a\bar{b}$. This inner product is called the **standard** inner product in F and it is the one assumed to equip F from now on.

1. When we choose F as the m th tensor power of F with the field multiplication as the tensor multiplication, then the canonical inner product is the inner product induced in $\bigotimes^m F$ by the canonical inner product.
2. When we assume V as the tensor product of F and V with the tensor multiplication $a \otimes \mathbf{v} = a\mathbf{v}$, the inner product induced by the canonical inner product of F and the inner product of V is the inner product of V .

3. Consider $L(V; U)$ as the tensor product of U and V^* by the tensor multiplication $(\mathbf{u} \otimes f)(\mathbf{v}) = f(\mathbf{v})\mathbf{u}$. Assume in V^* the inner product dual of the inner product of V . Then, if $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ is an orthonormal basis of V and $\theta, \eta \in L(V; U)$, we have

$$\langle \theta, \eta \rangle = \sum_{j=1}^m \langle \theta(\mathbf{v}_j), \eta(\mathbf{v}_j) \rangle = \text{tr}(\eta^* \theta),$$

i.e., the associated inner product of $L(V; U)$ is the Hilbert–Schmidt one.

4. Consider $F^{m \times n}$ as the tensor product of F^m and F^n by the tensor multiplication described in Example 1 in section 13.2. Then if we consider in F^m and F^n the usual inner product we get in $F^{m \times n}$ as the induced inner product, the inner product

$$(A, B) \mapsto \text{tr}(\overline{B}^T A) = \sum_{i,j} a_{ij} \overline{b_{i,j}}.$$

5. Assume that in V_i^* is defined the inner product dual of $\langle \cdot, \cdot \rangle_{V_i}$, $i = 1, \dots, m$. Then choosing $L(V_1, \dots, V_m; F)$ as the tensor product of $V_1^*, \dots, V_m^*$, with the usual tensor multiplication, the inner product of $L(V_1, \dots, V_m; F)$ induced by the duals of inner products on V_i^* , $i = 1, \dots, m$ is given by the equalities

$$\langle \varphi, \psi \rangle = \sum_{\alpha \in \Gamma} \varphi(\mathbf{b}_{1,\alpha(1)}, \dots, \mathbf{b}_{m,\alpha(m)}) \overline{\psi(\mathbf{b}_{1,\alpha(1)}, \dots, \mathbf{b}_{m,\alpha(m)})}.$$

13.11 Orientation and Hodge Star Operator

In this section, we assume that all vector spaces are real finite dimensional inner product spaces.

Definitions:

Let V be a one-dimensional vector space. The equivalence classes of the equivalence relation $\sim$, defined by the condition $\mathbf{v} \sim \mathbf{v}'$ if there exists a positive real number $a > 0$ such that $\mathbf{v}' = a\mathbf{v}$, partitions the set of nonzero vectors of V into two subsets.

Each one of these subsets is known as an open **half-line**.

An **orientation** of V is a choice of one of these subsets. The fixed open half-line is called the **positive half-line** and its vectors are known as **positive**. The other open half-line of V is called **negative half-line**, and its vectors are also called **negative**.

The field $\mathbb{R}$, regarded as one-dimensional vector space, has a “natural” orientation that corresponds to choose as positive half-line the set of positive numbers.

If V is an n -dimensional vector space, $\bigwedge^n V$ is a one-dimensional vector space (recall Fact 22 in section 13.6). An **orientation** of V is an orientation of $\bigwedge^n V$.

A basis $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ of V is said to be **positively oriented** if $\mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_n$ is positive and **negatively oriented** if $\mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_n$ is negative.

Throughout this section $\bigwedge^m V$ will be equipped with the inner product $\langle \cdot, \cdot \rangle_\wedge$, a positive multiple of the induced the inner product, defined by

$$\langle z, w \rangle_\wedge = \frac{1}{m!} \langle z, w \rangle,$$

where the inner product on the right-hand side of the former equality is the inner product of $\bigotimes^m V$ induced by the inner product of V . This is also the inner product that is considered whenever the norm of antisymmetric tensors is referred.

The positive tensor of norm 1 of $\bigwedge^n V, u_V$, is called **fundamental tensor of V** or **element of volume of V** .

Let V be a real oriented inner product space. Let $0 \leq m \leq n$.

The **Hodge star operator** is the linear operator $\star_m$ (denoted also by $\star$) from $\bigwedge^m V$ into $\bigwedge^{n-m} V$ defined by the following condition:

$$\langle \star_m(w), w' \rangle_{\bigwedge^{n-m} V} = w \wedge w', \text{ for all } w' \in \bigwedge^{n-m} V.$$

Let $n \geq 1$ and let V be an n -dimensional oriented inner product space over $\mathbb{R}$. The **external product on V** is the map

$$(\mathbf{v}_1, \dots, \mathbf{v}_{n-1}) \mapsto \mathbf{v}_1 \times \dots \times \mathbf{v}_{n-1} = \star_{n-1}(\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_{n-1}),$$

from V^{n-1} into V .

Facts:

The following facts can be found in [Mar75, Chap. 1] and [Sch75, Chap. 1].

1. If $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ is a positively oriented orthonormal basis of V , then $u_V = \mathbf{b}_1 \wedge \dots \wedge \mathbf{b}_n$.
2. If $(\mathbf{b}_1, \dots, \mathbf{b}_n)$ is a positively oriented orthonormal basis of V , then

$$\star_m \mathbf{b}_\alpha^\wedge = \text{sgn}(\tilde{\alpha}) \mathbf{b}_{\alpha^c}^\wedge, \quad \alpha \in Q_{m,n},$$

where $\tilde{\alpha}$ and α^c are defined in Section 13.7.

3. Let $(\mathbf{v}_1, \dots, \mathbf{v}_n)$ and $(\mathbf{u}_1, \dots, \mathbf{u}_n)$ be two bases of V and $\mathbf{v}_j = \sum a_{ij} \mathbf{u}_i$, $j = 1, \dots, n$. Let $A = [a_{ij}]$. Since (recall Fact 29 in Section 13.6)

$$\mathbf{v}_1 \wedge \dots \wedge \mathbf{v}_n = \det(A) \mathbf{u}_1 \wedge \dots \wedge \mathbf{u}_n,$$

two bases have the same orientation if and only if their transition matrix has a positive determinant.

4. $\star$ is an isometric isomorphism.
5. $\star_0$ is the linear isomorphism that maps $1 \in \mathbb{R}$ onto the fundamental tensor.
6. $\star_m \star_{n-m} = (-1)^{m(n-m)} I_{\bigwedge^{n-m} V}$.

Let V be an n -dimensional oriented inner product space over $\mathbb{R}$.

7. If $m \neq 0$ and $m \neq n$, the Hodge star operator maps the set of decomposable elements of $\bigwedge^m V$ onto the set of decomposable elements of $\bigwedge^{n-m} V$.
8. Let $(\mathbf{x}_1, \dots, \mathbf{x}_m)$ be a linearly independent family of vectors of V . Then

$$\mathbf{y}_1 \wedge \dots \wedge \mathbf{y}_{n-m} = \star_m(\mathbf{x}_1 \wedge \dots \wedge \mathbf{x}_m)$$

if and only if the following three conditions hold:

- (a) $\mathbf{y}_1, \dots, \mathbf{y}_{n-m} \in \text{Span}(\{\mathbf{x}_1, \dots, \mathbf{x}_m\})^\perp$;
- (b) $\|\mathbf{y}_1 \wedge \dots \wedge \mathbf{y}_{n-m}\| = \|\mathbf{x}_1 \wedge \dots \wedge \mathbf{x}_m\|$;
- (c) $(\mathbf{x}_1, \dots, \mathbf{x}_m, \mathbf{y}_1, \dots, \mathbf{y}_{n-m})$ is a positively oriented basis of V .

9. If $(\mathbf{v}_1, \dots, \mathbf{v}_{n-1})$ is linearly independent, $\mathbf{v}_1 \times \cdots \times \mathbf{v}_{n-1}$ is completely characterized by the following three conditions:

- (a) $\mathbf{v}_1 \times \cdots \times \mathbf{v}_{n-1} \in \text{Span}(\{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}\})^\perp$.
- (b) $\|\mathbf{v}_1 \times \cdots \times \mathbf{v}_{n-1}\| = \|\mathbf{v}_1 \wedge \cdots \wedge \mathbf{v}_{n-1}\|$.
- (c) $(\mathbf{v}_1, \dots, \mathbf{v}_{n-1}, \mathbf{v}_1 \times \cdots \times \mathbf{v}_{n-1})$ is a positively oriented basis of V .

10. Assume $V^* = L(V; F)$, with $\dim(V) \geq 1$, is equipped with the dual inner product. Consider $L^m(V; F)$ as a model for the m th tensor power of V^* with the usual tensor multiplication. Then $\bigwedge^m V^* = A^m(V; F)$. If λ is an antisymmetric form in $A^m(V; F)$, then $\star_m(\lambda)$ is the form whose value in $(\mathbf{v}_1, \dots, \mathbf{v}_{n-m})$ is the component in the fundamental tensor of $\lambda \wedge \varrho^{-1}(\mathbf{v}_1) \wedge \cdots \wedge \varrho^{-1}(\mathbf{v}_{n-m})$, where ϱ is defined in the definition of section 13.10.

$$\star_m(\lambda)(\mathbf{v}_1, \dots, \mathbf{v}_{n-m})u_{V^*} = \lambda \wedge \varrho^{-1}(\mathbf{v}_1) \wedge \cdots \wedge \varrho^{-1}(\mathbf{v}_{n-m}).$$

11. Assuming the above setting for the Hodge star operator, the external product of $\mathbf{v}_1, \dots, \mathbf{v}_{n-1}$ is the image by ϱ of the form $(u_{V^*})_{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}}$ (recall that $(u_{V^*})_{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}}(\mathbf{v}_n) = u_{V^*}(\mathbf{v}_1, \dots, \mathbf{v}_{n-1}, \mathbf{v}_n)$), i.e.,

$$\mathbf{v}_1 \times \cdots \times \mathbf{v}_{n-1} = \varrho((u_{V^*})_{\mathbf{v}_1, \dots, \mathbf{v}_{n-1}}).$$

The preceding formula can be unfolded by stating that for each $v \in V$, $\langle \mathbf{v}, \mathbf{v}_1 \times \cdots \times \mathbf{v}_{n-1} \rangle = u_{V^*}(\mathbf{v}_1, \dots, \mathbf{v}_{n-1}, \mathbf{v})$.

Examples:

1. If V has dimension 0, the isomorphism $\star_0$ from $\bigwedge^0 V = \mathbb{R}$ into $\bigwedge^0 V = \mathbb{R}$ is either the identity (in the case we choose the natural orientation of V) or $-I$ (in the case we fix the nonnatural orientation of V).
2. When V has dimension 2, the isomorphism $\star_1$ is usually denoted by J . It has the property $J^2 = -I$ and corresponds to the positively oriented rotation of $\pi/2$.
3. Assume that V has dimension 2. Then the external product is the isomorphism J .
4. If $\dim(V) = 3$, the external product is the well-known cross product.

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14

Matrix Equalities and Inequalities

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14.1	Eigenvalue Equalities and Inequalities	14-1
14.2	Spectrum Localization	14-5
14.3	Inequalities for the Singular Values and the Eigenvalues	14-8
14.4	Basic Determinantal Relations	14-10
14.5	Rank and Nullity Equalities and Inequalities	14-12
14.6	Useful Identities for the Inverse	14-15
	References	14-17

In this chapter, we have collected classical equalities and inequalities regarding the eigenvalues, the singular values, the determinant, and the dimensions of the fundamental subspaces of a matrix. Also included is a section on identities for matrix inverses. The majority of these results can be found in comprehensive books on linear algebra and matrix theory, although some are of specialized nature. The reader is encouraged to consult, e.g., [HJ85], [HJ91], [MM92], or [Mey00] for details, proofs, and further bibliography.

14.1 Eigenvalue Equalities and Inequalities

The majority of the facts in this section concern general matrices; however, some classical and frequently used results on eigenvalues of Hermitian and positive definite matrices are also included. For the latter, see also Chapter 8 and [HJ85, Chap. 4]. Many of the definitions and some of the facts in this section are also given in Section 4.3.

Facts:

1. [HJ85, Chap. 1] Let $A \in F^{n \times n}$, where $F = \mathbb{C}$ or any algebraically closed field. Let $p_A(x) = \det(xI - A)$ be the characteristic polynomial of A , and $\lambda_1, \lambda_2, \dots, \lambda_n$ be the eigenvalues of A . Denote by $S_k(\lambda_1, \dots, \lambda_n)$ ($k = 1, 2, \dots, n$) the k th elementary symmetric function of the eigenvalues (here abbreviated $S_k(\lambda)$), and by $S_k(A)$ the sum of all $k \times k$ principal minors of A . Then

- The characteristic polynomial satisfies

$$\begin{aligned} p_A(x) &= (x - \lambda_1)(x - \lambda_2) \cdots (x - \lambda_n) \\ &= x^n - S_1(\lambda)x^{n-1} + S_2(\lambda)x^{n-2} + \cdots + (-1)^{n-1}S_{n-1}(\lambda)x + (-1)^n S_n(\lambda) \\ &= x^n - S_1(A)x^{n-1} + S_2(A)x^{n-2} + \cdots + (-1)^{n-1}S_{n-1}(A)x + (-1)^n S_n(A). \end{aligned}$$

- $S_k(\lambda) = S_k(\lambda_1, \dots, \lambda_n) = S_k(A) (k = 1, 2, \dots, n)$.
 - $\text{tr} A = S_1(A) = \sum_{i=1}^n a_{ii} = \sum_{i=1}^n \lambda_i$ and $\det A = S_n(A) = \prod_{i=1}^n \lambda_i$.
2. [HJ85, (1.2.13)] Let $A(i)$ be obtained from $A \in \mathbb{C}^{n \times n}$ by deleting row and column i . Then

$$\frac{d}{dx} p_A(x) = \sum_{i=1}^n p_{A(i)}(x).$$

Facts 3 to 9 are collected, together with historical commentary, proofs, and further references, in [MM92, Chap. III].

3. (Hirsch and Bendixson) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ and λ be an eigenvalue of A . Denote $B = [b_{ij}] = (A + A^*)/2$ and $C = [c_{ij}] = (A - A^*)/(2i)$. Then the following inequalities hold:

$$\begin{aligned} |\lambda| &\leq n \max_{i,j} |a_{ij}|, \\ |\text{Re} \lambda| &\leq n \max_{i,j} |b_{ij}|, \\ |\text{Im} \lambda| &\leq n \max_{i,j} |c_{ij}|. \end{aligned}$$

Moreover, if $A + A^T \in \mathbb{R}^{n \times n}$, then

$$|\text{Im} \lambda| \leq \max_{i,j} |c_{ij}| \sqrt{\frac{n(n-1)}{2}}.$$

4. (Pick's inequality) Let $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ and λ be an eigenvalue of A . Denote $C = [c_{ij}] = (A - A^T)/2$. Then

$$|\text{Im} \lambda| \leq \max_{i,j} |c_{ij}| \cot\left(\frac{\pi}{2n}\right).$$

5. Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ and λ be an eigenvalue of A . Denote $B = [b_{ij}] = (A + A^*)/2$ and $C = [c_{ij}] = (A - A^*)/(2i)$. Then the following inequalities hold:

$$\begin{aligned} \min\{\mu : \mu \in \sigma(B)\} &\leq \text{Re} \lambda \leq \max\{\mu : \mu \in \sigma(B)\}, \\ \min\{\nu : \nu \in \sigma(C)\} &\leq \text{Im} \lambda \leq \max\{\nu : \nu \in \sigma(C)\}. \end{aligned}$$

6. (Schur's inequality) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ have eigenvalues λ_j ($j = 1, 2, \dots, n$). Then

$$\sum_{j=1}^n |\lambda_j|^2 \leq \sum_{i,j=1}^n |a_{ij}|^2$$

with equality holding if and only if A is a normal matrix (i.e., $A^* A = A A^*$). (See Section 7.2 for more information on normal matrices.)

7. (Browne's Theorem) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ and λ_j ($j = 1, 2, \dots, n$) be the eigenvalues of A ordered so that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$. Let also $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$ be the singular values of A , which are real and nonnegative. (See Section 5.6 for the definition.) Then

$$\sigma_n \leq |\lambda_j| \leq \sigma_1 \quad (j = 1, 2, \dots, n).$$

In fact, the following more general statement holds:

$$\prod_{i=1}^k \sigma_{n-i+1} \leq \prod_{j=1}^k |\lambda_{t_j}| \leq \prod_{i=1}^k \sigma_i,$$

for every $k \in \{1, 2, \dots, n\}$ and every k -tuple $(t_1, t_2, \dots, t_k)$ of strictly increasing elements chosen from $\{1, 2, \dots, n\}$.

8. Let $A \in \mathbb{C}^{n \times n}$ and R_i, C_i ($i = 1, 2, \dots, n$) denote the sums of the absolute values of the entries of A in row i and column i , respectively. Also denote

$$R = \max_i \{R_i\} \quad \text{and} \quad C = \max_i \{C_i\}.$$

Let λ be an eigenvalue of A . Then the following inequalities hold:

$$\begin{aligned} |\lambda| &\leq \max_i \frac{R_i + C_i}{2} \leq \frac{R + C}{2}, \\ |\lambda| &\leq \max_i \sqrt{R_i C_i} \leq \sqrt{RC}, \\ |\lambda| &\leq \min\{R, C\}. \end{aligned}$$

9. (Schneider's Theorem) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ and λ_j ($j = 1, 2, \dots, n$) be the eigenvalues of A ordered so that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$. Let $\mathbf{x} = [x_i]$ be any vector in $\mathbb{R}^n$ with positive entries and define the quantities

$$r_i = \sum_{j=1}^n \frac{|a_{ij}|x_j}{x_i} \quad (i = 1, 2, \dots, n).$$

Then

$$\prod_{j=1}^k |\lambda_j| \leq \prod_{j=1}^k r_{i_j} \quad (k = 1, 2, \dots, n)$$

for all n -tuples $(i_1, i_2, \dots, i_n)$ of elements from $\{1, 2, \dots, n\}$ such that

$$r_{i_1} \geq r_{i_2} \geq \dots \geq r_{i_n}.$$

10. [HJ85, Theorem 8.1.18] For $A = [a_{ij}] \in \mathbb{C}^{n \times n}$, let its entrywise absolute value be denoted by $|A| = [|a_{ij}|]$. Let $B \in \mathbb{C}^{n \times n}$ and assume that $|A| \leq B$ (entrywise). Then

$$\rho(A) \leq \rho(|A|) \leq \rho(B).$$

11. [HJ85, Chap. 5, Sec. 6] Let $A \in \mathbb{C}^{n \times n}$ and $\|\cdot\|$ denote any matrix norm on $\mathbb{C}^{n \times n}$. (See Chapter 37). Then

$$\rho(A) \leq \|A\|$$

and

$$\lim_{k \rightarrow \infty} \|A^k\|^{1/k} = \rho(A).$$

12. [HJ91, Corollary 1.5.5] Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$. The numerical range of $A \in \mathbb{C}^{n \times n}$ is $W(A) = \{v^* A v \in \mathbb{C} : v \in \mathbb{C}^n \text{ with } v^* v = 1\}$ and the numerical radius of $A \in \mathbb{C}^{n \times n}$ is $r(A) = \max\{|z| : z \in W(A)\}$. (See Chapter 18 for more information about the numerical range and numerical radius.) Then the following inequalities hold:

$$\begin{aligned} r(A^m) &\leq [r(A)]^m \quad (m = 1, 2, \dots), \\ \rho(A) &\leq r(A) \leq \frac{\|A\|_1 + \|A\|_\infty}{2}, \\ \frac{\|A\|_2}{2} &\leq r(A) \leq \|A\|_2, \\ r(A) &\leq r(|A|) = \frac{|A| + |A|^T}{2} \quad (\text{where } |A| = [|a_{ij}|]). \end{aligned}$$

Moreover, the following statements are equivalent:

- (a) $r(A) = \|A\|_2$.
- (b) $\rho(A) = \|A\|_2$.
- (c) $\|A^n\|_2 = \|A\|_2^n$.
- (d) $\|A^k\|_2 = \|A\|_2^k$ ($k = 1, 2, \dots$).

Facts 13 to 15 below, along with proofs, can be found in [HJ85, Chap. 4].

13. (Rayleigh–Ritz) Let $A \in \mathbb{C}^{n \times n}$ be Hermitian (i.e., $A = A^*$) with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Then
- (a) $\lambda_n x^* x \leq x^* A x \leq \lambda_1 x^* x$ for all $x \in \mathbb{C}^n$.
 - (b) $\lambda_1 = \max_{x \neq 0} \frac{x^* A x}{x^* x} = \max_{x^* x = 1} x^* A x$.
 - (c) $\lambda_n = \min_{x \neq 0} \frac{x^* A x}{x^* x} = \min_{x^* x = 1} x^* A x$.
14. (Courant–Fischer) Let $A \in \mathbb{C}^{n \times n}$ be Hermitian with eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$. Let $k \in \{1, 2, \dots, n\}$. Then

$$\begin{aligned} \lambda_k &= \min_{W_1, W_2, \dots, W_{k-1} \in \mathbb{C}^n} \max_{\substack{x \neq 0, x \in \mathbb{C}^n \\ x \perp W_1, W_2, \dots, W_{k-1}}} \frac{x^* A x}{x^* x} \\ &= \max_{W_1, W_2, \dots, W_{n-k} \in \mathbb{C}^n} \min_{\substack{x \neq 0, x \in \mathbb{C}^n \\ x \perp W_1, W_2, \dots, W_{n-k}}} \frac{x^* A x}{x^* x}. \end{aligned}$$

15. (Weyl) Let $A, B \in \mathbb{C}^{n \times n}$ be Hermitian. Consider the eigenvalues of A , B , and $A + B$, denoted by $\lambda_i(A)$, $\lambda_i(B)$, $\lambda_i(A + B)$, respectively, arranged in decreasing order. Then the following hold:

- (a) For each $k \in \{1, 2, \dots, n\}$,

$$\lambda_k(A) + \lambda_n(A) \leq \lambda_k(A + B) \leq \lambda_k(A) + \lambda_1(B).$$

- (b) For every pair $j, k \in \{1, 2, \dots, n\}$ such that $j + k \geq n + 1$,

$$\lambda_{j+k-n}(A + B) \geq \lambda_j(A) + \lambda_k(B).$$

- (c) For every pair $j, k \in \{1, 2, \dots, n\}$ such that $j + k \leq n + 1$,

$$\lambda_j(A) + \lambda_k(B) \geq \lambda_{j+k-1}(A + B).$$

Examples:

1. To illustrate several of the facts in this section, consider

$$A = \begin{bmatrix} 1 & -1 & 0 & 2 \\ 3 & 1 & -2 & 1 \\ 1 & 0 & 0 & -1 \\ -1 & 2 & 1 & 0 \end{bmatrix},$$

whose spectrum, $\sigma(A)$, consists of

$$\lambda_1 = -0.7112 + 2.6718i, \lambda_2 = -0.7112 - 2.6718i, \lambda_3 = 2.5506, \lambda_4 = 0.8719.$$

Note that the eigenvalues are ordered decreasingly with respect to their moduli (absolute values):

$$|\lambda_1| = |\lambda_2| = 2.7649 > |\lambda_3| = 2.5506 > |\lambda_4| = 0.8719.$$

The maximum and minimum eigenvalues of $(A + A^*)/2$ are 2.8484 and -1.495 . Note that, as required by Fact 5, for every $\lambda \in \sigma(A)$,

$$-1.495 \leq |\lambda| \leq 2.8484.$$

To illustrate Fact 7, let $(t_1, t_2) = (1, 3)$ and compute the singular values of A :

$$\sigma_1 = 4.2418, \sigma_2 = 2.5334, \sigma_3 = 1.9890, \sigma_4 = 0.7954.$$

Then, indeed,

$$\sigma_4 \sigma_3 = 1.5821 \leq |\lambda_1| |\lambda_3| = 7.0522 \leq \sigma_1 \sigma_2 = 10.7462.$$

Referring to the notation in Fact 8, we have $C = 6$ and $R = 7$. The spectral radius of A is $\rho(A) = 2.7649$ and, thus, the modulus of every eigenvalue of A is indeed bounded above by the quantities

$$\frac{R+C}{2} = \frac{13}{2} = 6.5, \quad \sqrt{RC} = 6.4807, \quad \min\{R, C\} = 6.$$

Letting B denote the entrywise absolute value of A , Facts 10 and 11 state that

$$\rho(A) = 2.7649 \leq \rho(B) = 4.4005 \quad \text{and} \quad \rho(A) = 2.7649 \leq \|A\|_2 = 4.2418.$$

Examples related to Fact 12 and the numerical range are found in Chapter 18. See also Example 2 that associates the numerical range with the location of the eigenvalues.

2. Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

and note that for every integer $m \geq 2$, A^m consists of zero entries, except for its $(1, 1)$ entry that is equal to 1. One may easily verify that

$$\rho(A) = 1, \quad \|A\|_\infty = \|A\|_1 = \|A\|_2 = 1.$$

By Fact 12, it follows that $r(A) = 1$ and all of the equivalent conditions (a) to (d) in that fact hold, despite A not being a normal matrix.

14.2 Spectrum Localization

This section presents results on classical inclusion regions for the eigenvalues of a matrix. The following facts, proofs, and details, as well as additional references, can be found in [MM92, Chap. III, Sec. 2], [HJ85, Chap. 6], and [Bru82].

Facts:

1. (Geršgorin) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ and define the quantities

$$R_i = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \quad (i = 1, 2, \dots, n).$$

Consider the **Geršgorin discs** (centered at a_{ii} with radii R_i),

$$D_i = \{z \in \mathbb{C} : |z - a_{ii}| \leq R_i\} \quad (i = 1, 2, \dots, n).$$

Then all the eigenvalues of A lie in the union of the Geršgorin discs; that is,

$$\sigma(A) \subset \bigcup_{i=1}^n D_i.$$

Moreover, if the union of k Geršgorin discs, G , forms a connected region disjoint from the remaining $n - k$ discs, then G contains exactly k eigenvalues of A (counting algebraic multiplicities).

2. (Lévy–Desplanques) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ be a strictly diagonally dominant matrix, namely,

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \quad (i = 1, 2, \dots, n).$$

Then A is an invertible matrix.

3. (Brauer) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ and define the quantities

$$R_i = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \quad (i = 1, 2, \dots, n).$$

Consider the **ovals of Cassini**, which are defined by

$$V_{i,j} = \{z \in \mathbb{C} : |z - a_{ii}||z - a_{jj}| \leq R_i R_j\} \quad (i, j = 1, 2, \dots, n, i \neq j).$$

Then all the eigenvalues of A lie in the union of the ovals of Cassini; that is,

$$\sigma(A) \subset \bigcup_{\substack{i,j=1 \\ i \neq j}}^n V_{i,j}.$$

4. [VK99, Eq. 3.1] Denoting the union of the Geršgorin discs of $A \in \mathbb{C}^{n \times n}$ by $\Gamma(A)$ (see Fact 1) and the union of the ovals of Cassini of A by $K(A)$ (see Fact 2), we have that

$$\sigma(A) \subset K(A) \subseteq \Gamma(A).$$

That is, the ovals of Cassini provided at least as good a localization for the eigenvalues of A as do the Geršgorin discs.

5. Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ such that

$$|a_{ii}||a_{kk}| > \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \sum_{\substack{j=1 \\ j \neq k}}^n |a_{kj}| \quad (i, k = 1, 2, \dots, n, i \neq k).$$

Then A is an invertible matrix.

6. Facts 1 to 5 can also be stated in terms of *column sums* instead of row sums.
7. (Ostrowski) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ and $\alpha \in [0, 1]$. Define the quantities

$$R_i = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}|, \quad C_i = \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ji}| \quad (i = 1, 2, \dots, n).$$

Then all the eigenvalues of A lie in the union of the discs

$$D_i(\alpha) = \{z \in \mathbb{C} : |z - a_{ii}| \leq R_i^\alpha C_i^{1-\alpha}\} \quad (i = 1, 2, \dots, n);$$

that is,

$$\sigma(A) \subset \bigcup_{i=1}^n D_i(\alpha).$$

8. Let $A \in \mathbb{C}^{n \times n}$ and consider the spectrum of A , $\sigma(A)$, as well as its numerical range, $W(A)$. Then

$$\sigma(A) \subset W(A).$$

In particular, if A is a normal matrix (i.e., $A^*A = AA^*$), then $W(A)$ is exactly equal to the convex hull of the eigenvalues of A .

Examples:

1. To illustrate Fact 1 (see also Facts 3 and 4) let

$$A = \begin{bmatrix} 3i & 1 & 0.5 & -1 & 0 \\ -1 & 2i & 1.5 & 0 & 0 \\ 1 & 2 & -7 & 0 & 1 \\ 0 & -1 & 0 & 10 & i \\ 1 & 0 & 1 & -1 & 1 \end{bmatrix}$$

and consider the Geršgorin discs of A displayed in Figure 14.1. Note that there are three connected regions of discs that are disjoint of each other. Each region contains as many eigenvalues (marked with +’s) as the number of discs it comprises. The ovals of Cassini are contained in the union of the Geršgorin discs. In general, although it is easy to verify whether a complex number belongs to an oval of Cassini or not, these ovals are generally difficult to draw. An interactive supplement to [VK99] (accessible at: www.emis.math.ca/EMIS/journals/ETNA/vol.8.1999/pp15-20.dir/gershini.html) allows one to draw and compare the Geršgorin discs and ovals of Cassini of 3×3 matrices.

2. To illustrate Fact 8, consider the matrices

$$A = \begin{bmatrix} 1 & -1 & 2 \\ 2 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2+2i & -2-i & -1-2i \\ 1+2i & -1-i & -1-2i \\ 2+i & -2-i & -1-i \end{bmatrix}.$$

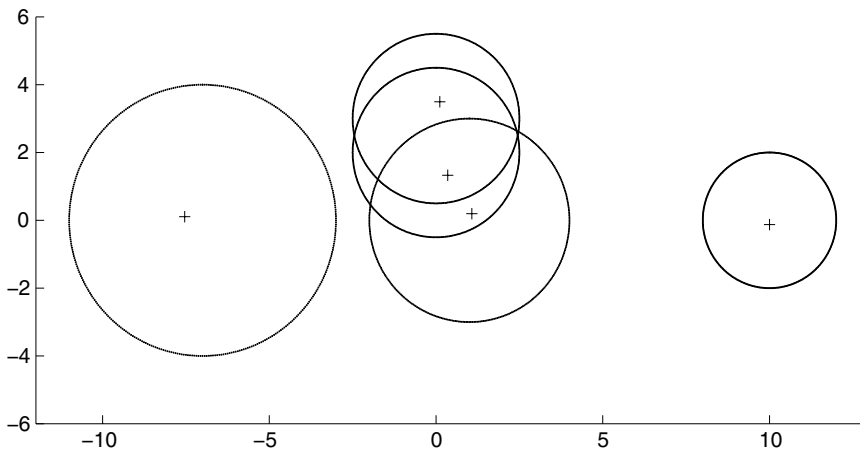


FIGURE 14.1 The Geršgorin disks of A .

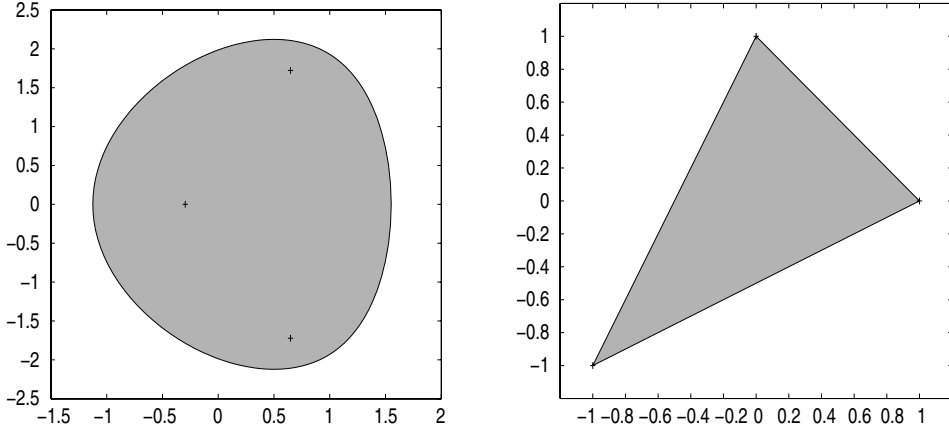


FIGURE 14.2 The numerical range of A and of the normal matrix B .

Note that B is a normal matrix with spectrum $\{1, i, -1 - i\}$. As indicated in Figure 14.2, the numerical ranges of A and B contain the eigenvalues of A and B , respectively, marked with '+'s. The numerical range of B is indeed the convex hull of the eigenvalues.

14.3 Inequalities for the Singular Values and the Eigenvalues

The material in this section is a selection of classical inequalities about the singular values. Extensive details and proofs, as well as a host of additional results on singular values, can be found in [HJ91, Chap. 3]. Definitions of many of the terms in this section are given in Section 5.6, Chapter 17, and Chapter 45; additional facts and examples are also given there.

Facts:

1. Let $A \in \mathbb{C}^{m \times n}$ and σ_1 be its largest singular value. Then

$$\sigma_1 = \|A\|_2.$$

2. Let $A \in \mathbb{C}^{m \times n}$, $q = \min\{m, n\}$. Denote the singular values of A by $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_q$ and let $k \in \{1, 2, \dots, q\}$. Then

$$\begin{aligned} \sigma_k &= \min_{W_1, W_2, \dots, W_{k-1} \in \mathbb{C}^n} \max_{\substack{\|x\|_2=1, x \in \mathbb{C}^n \\ x \perp W_1, W_2, \dots, W_{k-1}}} \|Ax\|_2 \\ &= \max_{W_1, W_2, \dots, W_{n-k} \in \mathbb{C}^n} \min_{\substack{\|x\|_2=1, x \in \mathbb{C}^n \\ x \perp W_1, W_2, \dots, W_{n-k}}} \|Ax\|_2 \\ &= \min_{\substack{W \subseteq \mathbb{C}^n \\ \dim W = n-k+1}} \max_{\substack{x \in W \\ \|x\|_2=1}} \|Ax\|_2 \\ &= \max_{\substack{W \subseteq \mathbb{C}^n \\ \dim W = k}} \min_{\substack{x \in W \\ \|x\|_2=1}} \|Ax\|_2, \end{aligned}$$

where the optimizations take place over all subspaces $W \subseteq \mathbb{C}^n$ of the indicated dimensions.

3. (Weyl) Let $A \in \mathbb{C}^{n \times n}$ have singular values $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_n$ and eigenvalues λ_j ($j = 1, 2, \dots, n$) be ordered so that $|\lambda_1| \geq |\lambda_2| \geq \dots \geq |\lambda_n|$. Then

$$|\lambda_1 \lambda_2 \cdots \lambda_k| \leq \sigma_1 \sigma_2 \cdots \sigma_k \quad (k = 1, 2, \dots, n).$$

Equality holds in (3) when $k = n$.

4. (A. Horn) Let $A \in \mathbb{C}^{m \times p}$ and $B \in \mathbb{C}^{p \times n}$. Let also $r = \min\{m, p\}$, $s = \min\{p, n\}$, and $q = \min\{r, s\}$. Denote the singular values of A , B , and AB , respectively, by $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$, $\tau_1 \geq \tau_2 \geq \cdots \geq \tau_s$, and $\chi_1 \geq \chi_2 \geq \cdots \geq \chi_q$. Then

$$\prod_{i=1}^k \chi_i \leq \prod_{i=1}^k \sigma_i \tau_i \quad (k = 1, 2, \dots, q).$$

Equality holds if $k = n = p = m$. Also for any $t > 0$,

$$\sum_{i=1}^k \chi_i^t \leq \sum_{i=1}^k (\sigma_i \tau_i)^t \quad (k = 1, 2, \dots, q).$$

5. Let $A \in \mathbb{C}^{n \times n}$ have singular values $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_n$ and eigenvalues λ_j ($j = 1, 2, \dots, n$) ordered so that $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|$. Then for any $t > 0$,

$$\sum_{i=1}^k |\lambda_i|^t \leq \sum_{i=1}^k \sigma_i^t \quad (k = 1, 2, \dots, n).$$

In particular, for $t = 1$ and $k = n$ we obtain from the inequality above that

$$|\operatorname{tr} A| \leq \sum_{i=1}^n \sigma_i.$$

6. Let $A, B \in \mathbb{C}^{m \times n}$ and $q = \min\{m, n\}$. Denote the singular values of A , B , and $A + B$, respectively, by $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_q$, $\tau_1 \geq \tau_2 \geq \cdots \geq \tau_q$, and $\psi_1 \geq \psi_2 \geq \cdots \geq \psi_q$. Then the following inequalities hold:

(a) $\psi_{i+j-1} \leq \sigma_i + \tau_j \quad (1 \leq i, j \leq q, \quad i + j \leq q + 1).$

(b) $|\rho_i - \sigma_i| \leq \tau_1 \quad (i = 1, 2, \dots, q).$

(c) $\sum_{i=1}^k \psi_i \leq \sum_{i=1}^k \sigma_i + \sum_{i=1}^k \tau_i \quad (k = 1, 2, \dots, q).$

7. Let $A \in \mathbb{C}^{n \times n}$ have eigenvalues λ_j ($j = 1, 2, \dots, n$) ordered so that $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|$. Denote the singular values of A^k by $\sigma_1(A^k) \geq \sigma_2(A^k) \geq \cdots \geq \sigma_n(A^k)$. Then

$$\lim_{k \rightarrow \infty} [\sigma_i(A^k)]^{1/k} = |\lambda_i| \quad (i = 1, 2, \dots, n).$$

Examples:

1. To illustrate Facts 1, 3, and 5, as well as gauge the bounds they provide, let

$$A = \begin{bmatrix} i & 2 & -1 & 0 \\ 2 & 1+i & 1 & 0 \\ 2i & 1 & 1 & 0 \\ 0 & 1-i & 1 & 0 \end{bmatrix},$$

whose eigenvalues and singular values ordered as required in Fact 3 are, respectively,

$$\lambda_1 = 2.6775 + 1.0227i, \lambda_2 = -2.0773 + 1.4685i, \lambda_3 = 1.3998 - 0.4912i, \lambda_4 = 0,$$

and

$$\sigma_1 = 3.5278, \sigma_2 = 2.5360, \sigma_3 = 1.7673, \sigma_4 = 0.$$

According to Fact 1, $\|A\|_2 = \sigma_1 = 3.5278$. The following inequalities hold according to Fact 3:

$$7.2914 = |\lambda_1 \lambda_2| \leq \sigma_1 \sigma_2 = 8.9465.$$

$$10.8167 = |\lambda_1 \lambda_2 \lambda_3| \leq \sigma_1 \sigma_2 \sigma_3 = 15.8114.$$

Finally, applying Fact 5 with $t = 3/2$ and $k = 2$, we obtain the inequality

$$8.9099 = |\lambda_1|^{3/2} + |\lambda_2|^{3/2} \leq \sigma_1^{3/2} + \sigma_2^{3/2} = 10.6646.$$

For $t = 1$ and $k = n$, we get

$$2.8284 = |2 + 2i| = |\text{tr}(A)| \leq \sum_{j=1}^4 \sigma_j = 7.8311.$$

14.4 Basic Determinantal Relations

The purpose of this section is to review some basic equalities and inequalities regarding the determinant of a matrix. For most of the facts mentioned here, see [Mey00, Chap. 6] and [HJ85, Chap. 0]. Definitions of many of the terms in this section are given in Sections 4.1 and 4.2; additional facts and examples are given there as well. Note that this section concludes with a couple of classical determinantal inequalities for positive semidefinite matrices; see Section 8.4 or [HJ85, Chap. 7] for more on this subject.

Following are some of the properties of determinants of $n \times n$ matrices, as well as classical formulas for the determinant of A and its submatrices.

Facts:

- Let $A \in F^{n \times n}$. The following are basic facts about the determinant. (See also Chapter 4.1.)
 - $\det A = \det A^T$; if $F = \mathbb{C}$, then $\det A^* = \overline{\det A}$.
 - If B is obtained from A by multiplying one row (or column) by a scalar c , then $\det B = c \det A$.
 - $\det(cA) = c^n \det A$ for any scalar c .
 - $\det(AB) = \det A \det B$. If A is invertible, then $\det A^{-1} = (\det A)^{-1}$.
 - If B is obtained from A by adding nonzero multiples of one row (respectively, column) to other rows (respectively, columns), then $\det B = \det A$.
 - $\det A = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$, where the summation is taken over all permutations σ of n letters, and where $\text{sgn}(\sigma)$ denotes the sign of the permutation σ .
 - Let A_{ij} denote the $(n-1) \times (n-1)$ matrix obtained from $A \in F^{n \times n}$ ($n \geq 2$) by deleting row i and column j . The following formula is known as the **Laplace expansion of $\det A$ along column j** :

$$\det A = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det A_{ij} \quad (j = 1, 2, \dots, n).$$

- (Cauchy–Binet) Let $A \in F^{m,k}$, $B \in F^{k \times n}$ and consider the matrix $C = AB \in F^{m \times n}$. Let also $\alpha \subseteq \{1, 2, \dots, m\}$ and $\beta \subseteq \{1, 2, \dots, n\}$ have cardinality r , where $1 \leq r \leq \min\{m, k, n\}$. Then the submatrix of C whose rows are indexed by α and columns indexed by β satisfies

$$\det C[\alpha, \beta] = \sum_{\substack{\gamma \subseteq \{1, 2, \dots, k\} \\ |\gamma| = r}} \det A[\alpha, \gamma] \det B[\gamma, \beta].$$

- [Mey00, Sec. 6.1, p. 471] Let $A = [a_{ij}(x)]$ be an $n \times n$ matrix whose entries are complex differentiable functions of x . Let D_i ($i = 1, 2, \dots, n$) denote the $n \times n$ matrix obtained from A when the entries in its i th row are replaced by their derivatives with respect to x . Then

$$\frac{d}{dx}(\det A) = \sum_{i=1}^n \det D_i.$$

4. Let $A = [a_{ij}]$ be an $n \times n$ matrix and consider its entries as independent variables. Then

$$\frac{\partial(\det A)}{\partial a_{ij}} = \det A(\{i\}, \{j\}) \quad (i, j = 1, 2, \dots, n),$$

where $A(\{i\}, \{j\})$ denotes the submatrix of A obtained from A by deleting row i and column j .

5. [Mey00, Sec. 6.2] Let $A \in F^{n \times n}$ and $\alpha \subseteq \{1, 2, \dots, n\}$. If the submatrix of A whose rows and columns are indexed by α , $A[\alpha]$, is invertible, then

$$\det A = \det A[\alpha] \det(A/A[\alpha]).$$

In particular, if A is partitioned in blocks as

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix},$$

where A_{11} and A_{22} are square matrices, then

$$\det A = \begin{cases} \det A_{11} \det(A_{22} - A_{21}(A_{11})^{-1}A_{12}) & \text{if } A_{11} \text{ is invertible} \\ \det A_{22} \det(A_{11} - A_{12}(A_{22})^{-1}A_{21}) & \text{if } A_{22} \text{ is invertible.} \end{cases}$$

The following two facts for $F = \mathbb{C}$ can be found in [Mey00, Sec. 6.2, pp. 475, 483] and [Mey00, Exer. 6.2.15, p. 485], respectively. The proofs are valid for arbitrary fields.

6. Let $A \in F^{n \times n}$ be invertible and $c, d \in F^n$. Then

$$\det(A + cd^T) = \det(A)(1 + d^T A^{-1}c).$$

7. Let $A \in F^{n \times n}$ be invertible, $x, y \in F^n$. Then

$$\det \begin{bmatrix} A & x \\ y^T & -1 \end{bmatrix} = -\det(A + xy^T).$$

8. [HJ85, Theorem 7.8.1 and Corollary 7.8.2] (Hadamard's inequalities) Let $A = [a_{ij}] \in \mathbb{C}^{n \times n}$ be a positive semidefinite matrix. Then

$$\det A \leq \prod_{i=1}^n a_{ii}.$$

If A is positive definite, equality holds if and only if A is a diagonal matrix.

For a general matrix $B = [b_{ij}] \in \mathbb{C}^{n \times n}$, applying the above inequality to B^*B and BB^* , respectively, one obtains

$$|\det B| \leq \prod_{i=1}^n \left(\sum_{j=1}^n |b_{ij}|^2 \right)^{1/2} \quad \text{and} \quad |\det B| \leq \prod_{j=1}^n \left(\sum_{i=1}^n |b_{ij}|^2 \right)^{1/2}.$$

If B is nonsingular, equalities hold, respectively, if and only if the rows or the columns of B are orthogonal.

9. [HJ85, Theorem 7.8.3] (Fischer's inequality) Consider a positive definite matrix

$$A = \begin{bmatrix} X & Y \\ Y^* & Z \end{bmatrix},$$

partitioned so that X, Z are square and nonvacuous. Then

$$\det A \leq \det X \det Z.$$